ADVANCE TRAUMA

Clinical Trial Protocol

Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Stepped-Wedge Cluster Randomised Trial

Version 1.6.0, 2026-11-09

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1 Visual abstract

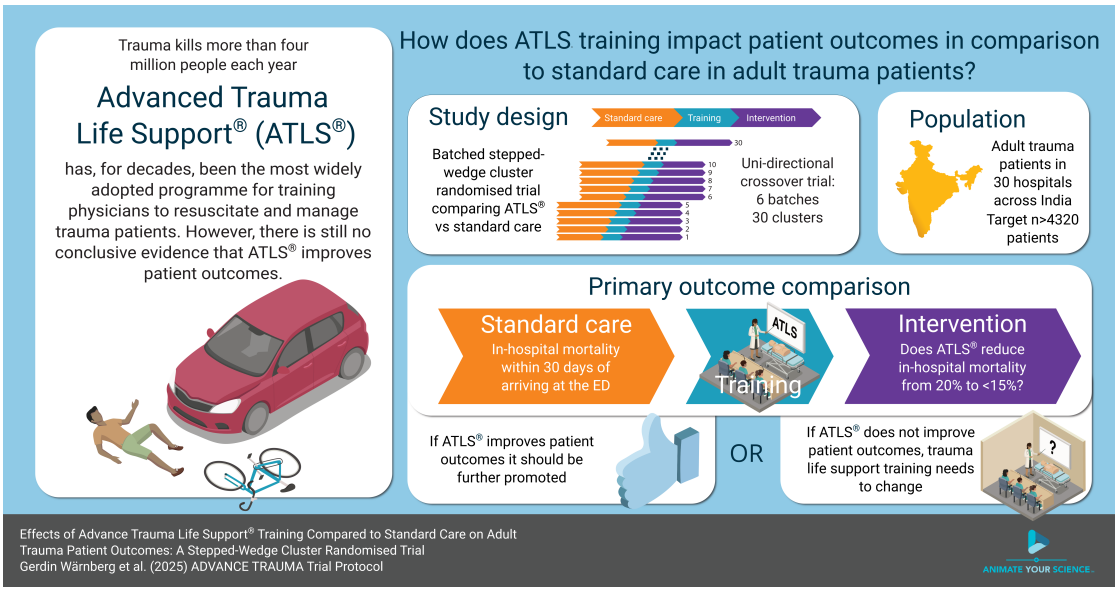


Figure 1: ADVANCE TRAUMA visual abstract.

2 Administrative information

2.1 Changelog

Version	Date	Details
1.6.0	2026-11-09	• Updated title page and text formatting • Updated visual abstract • Shortened very long lists of variable categories • Added references to published systematic review, feasibility study and community consultations pre-print • Removed note on bespoke code for randomisation • Replaced RSPL estimation (SAS-specific) with R alternative • Fixed grammar and spelling errors
1.5.0	2025-06-12	• Clarified the definition of clusters and units, based on discussions during the writeup of the protocol for publication • Clarified the patient exclusion criterion of isolated limb injury as isolated closed extremity fracture • Included “stepped-wedge” in the title • Corrected Eligibility criteria heading numbering • Added citation for the nested staircase design • Revise intention to treat terminology
1.4.0	2025-04-30	• Updated systematic review results and references • Added visual abstract • Added details on the nested staircase design for measuring adherence, quality of life and disability • Removed return to work as a secondary outcome to be measured using the nested staircase design, this should be measured using the main stepped-wedge design
1.3.0	2024-11-15	• Updated names of events in the table of procedures • Added new references • Added nested staircase design for measuring adherence, quality of life, disability and return to work • Updated small sample correction to be based on best available evidence closer to the time of analysis • Added contributors • Removed reassessment of the sample size calculation from the interim analysis • Revised details on measuring ATLS adherence

Version	Date	Details
1.2.0	2024-08-26	• Added details on measuring ATLS adherence • Clarified the section describing the consent process • Fixed minor issues with how the variables were listed • Indicated non-routinely recorded data in the list of variables • Added Administrative information section with contributors • Added CTRI registration number
1.1.0	2024-05-09	Updated the primary outcome to in-hospital mortality and spelling corrections. The primary outcome was updated following a voting procedure in the Trial Management Group.

2.2 Study identifiers

- ClinicalTrials.gov identifier: NCT06321419
- Clinical Trials Registry - India identifier: CTRI/2024/07/071336

2.3 Contributors

The following have contributed to the design and implementation of the trial:

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Abbreviations: TMG, Trial Management Group; TT, Trial Team.

3 Synopsis

Title Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Stepped-Wedge Cluster Randomised Trial

Rationale Trauma is a massive global health issue. Many training programmes have been developed to help physicians in the initial management of trauma patients. Among these programmes, Advanced Trauma Life Support® (ATLS®) is the most popular, having trained over one million physicians worldwide. Despite its widespread use, there are no controlled trials showing that ATLS® improves patient outcomes. Multiple systematic reviews emphasise the need for such trials.

Aim To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

Primary Outcome In-hospital mortality within 30 days of arrival at the emergency department.

Trial Design Batched stepped-wedge cluster randomised trial in India.

Trial Population Adult trauma patients presenting to the emergency department of a participating hospital.

Sample Size 30 clusters and at least 4320 patients.

Eligibility Criteria

Clusters are secondary or tertiary hospitals providing trauma care in India that admit or refer/transfer for admission at least 400 patients with trauma per year. Within each cluster, one or more units of physicians providing initial trauma care in the emergency department will be trained.

Participants are adult trauma patients who present to the emergency department of participating hospitals and are admitted or transferred for admission.

Intervention The intervention will be ATLS® training, a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. Physicians will be trained in an accredited ATLS® training facility in India.

Ethical Considerations We will use an opt-out consent approach for collection of routinely recorded data. We will obtain informed consent for collection of non-routinely recorded data, such as quality of life and disability outcomes. Patients who are unconscious or lack a legally authorised representative will be included under a waiver of informed consent. Note that consent here refers to consent to data collection.

Trial Period February 2025 to October 2029

4 Background and rationale

Each year, 4.9 million people die from trauma[1]. Among people aged 10-49 years trauma is the largest cause of disability adjusted life years[2]. Most deaths from trauma occur within the first 24-48 hours[3]. Traumatic brain injury and exsanguination are the most common causes of trauma deaths[4,5]. Most preventable trauma deaths are caused by clinical judgement errors during initial resuscitation or early care including airway management and haemorrhage control, even though the deaths occur later during the hospital stay[4,6].

Several trauma life support training programmes have been developed to improve the early management of patients in the hospital by providing a structured framework for assessment and treatment[7–11]. The proprietary Advanced Trauma Life Support® (ATLS®) is the most established trauma life support training programme and more than one million physicians in over 80 countries have been trained in the programme since the first course in 1978[12]. In the US and many other countries training in ATLS® is virtually mandatory for trauma care physicians[13]. Uptake in low- and middle income countries (LMIC) has been slow, potentially due to high costs[9].

There are three randomised studies showing that ATLS® improves knowledge and clinical skills[14–16], but there are no randomised controlled trials or high-quality quasi-experimental trials indicating that ATLS® improves patient outcomes[7,8,10,11,17,18]. We updated our published systematic review[18] and estimated a pooled odds ratio of 0.62 (95% CI 0.44; 0.89) from 12 heterogeneous (I^2 89.2%) observational studies on the effect of ATLS on mortality, see Updated systematic review (Section 4.1) for details[19–30].

We conducted a pilot cluster randomised controlled trial (ClinicalTrials.gov NCT05417243) between April 2022 and February 2023 as part of our network grant to assess the feasibility of a full scale trial[31,32]. Our pilot study enrolled 376 patients from seven hospitals across India and shows that it is feasible to conduct the proposed trial with a high percentage of patients consenting to out of hospital follow up (78%), low loss to follow-up rate (1%), and low missingness in key variables (mean 0.8%).

To involve patients and the public in the planning of this trial we conducted 19 semi-structured interviews with trauma patients, caregivers, and community representatives[33]. The aim of these interviews was to understand their views on the trial and important outcomes and the interviews showed high acceptability of our research and emphasised the importance of better recovery before discharge and functional outcomes at and after discharge, including pain, mobility and self-care activities. The interviews also highlighted return to work as an important outcome.

4.1 Updated systematic review

We perform systematic literature searches in collaboration with the Karolinska Institutet University Library, using Medline, Embase, Cochrane, Web of Science, Global Health, CINAHL and Google Scholar databases. We published our findings in 2026[18], based on a search in August 2025, but our latest search for this protocol was conducted in February 2026. We developed the search strategy in Medline (Ovid), limited the search to English language articles, searched all databases from inception, and screened a total of 10,046 records. We used a random effects model to pool estimates across studies. For this protocol, we pooled all 12 eligible ATLS® studies regardless of their quality rating. The published

review only included the ten studies deemed to be of sufficiently high quality, arriving at a pooled OR of 0.51 (96% CI 0.37-0.69)[18].

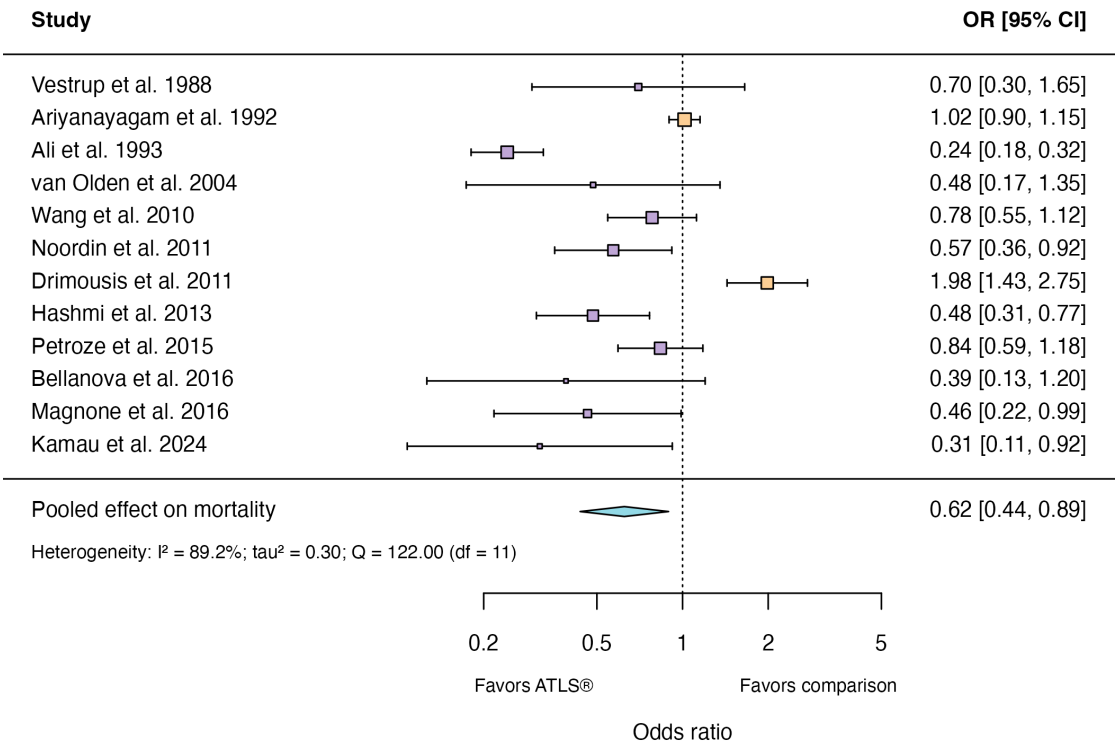


Figure 2: Summary of the updated systematic review. The forest plot shows the association between ATLS® and mortality in all eligible ATLS studies, including the two studies excluded from the published meta-analysis because of their quality ratings[18]. Abbreviations: OR, odds ratio; CI, confidence interval; ATLS®, Advanced Trauma Life Support; I^2, heterogeneity.

5 Benefit-risk evaluation

The direct risks includes integrity violations and data leakage. We will mitigate these risks by employing rigorous data collection and storage mechanisms. The procedures that we will use to collect data will be direct observation of care, routine physical examinations, questionnaires, and extraction of already collected data from patient records, which are often seen as involving only minimal risk.

The long-term risks of the research and the risk that the research will be used in detrimental ways are minimal. Our trial will assess the effect of Advanced Trauma Life Support® (ATLS®) on patient outcomes. Training in ATLS® is standard in many health care systems and it is unlikely that training physicians in this programme induces any harm to participants.

We consider these risks weighed up by the potential direct benefit for the participants in the intervention phase, if ATLS® is found to improve patient outcomes, and by the potential for improved care for the trauma patient population.

6 Trial aim

To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

7 Regulatory approvals and trial registration

We will submit this trial to the Health Ministry Screening Committee at the Indian Council for Medical Research for their approval. We will apply for ethical approvals from each participating hospital, The George Institute for Global Health in India and the Swedish Ethical Review Authority. We will register this trial with Clinical Trials Registry-India and ClinicalTrials.gov.

8 Trial design and procedures

8.1 Overall trial design

We will conduct a batched stepped-wedge cluster randomised controlled trial (see Figure 3). The stepped-wedge trial is a uni-directional cross-over trial but the time point when clusters cross-over from standard care to the intervention is randomised[34]. In this trial, the unit of randomisation is the hospital: that is, the "clusters" randomised to the sequences of the stepped wedge design are the hospitals. Within each hospital, one or more units of physicians performing initial resuscitation of trauma patients in the emergency department will be trained. The number of units that will be trained in each hospital will depend on the sizes of these units and the volumes of patients the physicians attend. If more than one unit is trained in the same hospital, these are all considered to be part of the same cluster (i.e. hospital) for the purpose of randomisation. We choose this approach for two reasons: 1) it will not be logistically or financially feasible to train all physician in a given hospital; and 2) we need to balance cluster size with the number of clusters. We will conduct this trial in India because physicians providing initial trauma care in India are so far not routinely trained in ATLS® or similar programmes.

We will roll out the interventions to 30 clusters over six batches, so there will be five clusters in each batch. The clusters in each batch will be randomised to one of five implementation sequences, with one hospital randomised to each implementation sequence. All clusters will transition through three phases, first a standard care phase, then a one month transition phase during which the training is delivered, and finally an intervention phase, for a total of 13 months. The implementation sequence determines how long the phases of standard care and intervention are. Patient participants will be followed up for a total of three months.

Within the main stepped-wedge design, we will nest a staircase design to measure a range of secondary outcomes (see Figure 4)[35]. The staircase design will include a random subset of patients presenting

during the three months preceding the transition phase, and the three months following the transition phase.

8.2 Design justification

We use the cluster randomised design because the intervention cannot be randomised at the individual patient level. We use the stepped-wedge design for two reasons. First, this design is statistically more efficient than the parallel cluster design when the number of clusters is limited[36]. In this trial, the number of clusters is limited because of the costs associated with ATLS® training and the available slots for ATLS® training in India. Second, the stepped-wedge design is likely to enhance participation and engagement because all clusters receive the intervention. The batched stepped-wedge design further improves feasibility as it does not require all clusters to start at the same time, and it is robust to potential delays in cluster recruitment[37]. We nest a staircase design within the main design because some of the secondary outcomes will be considerably more labour intensive to collect than the primary outcome, and collecting these for all patient participants would be unfeasible.

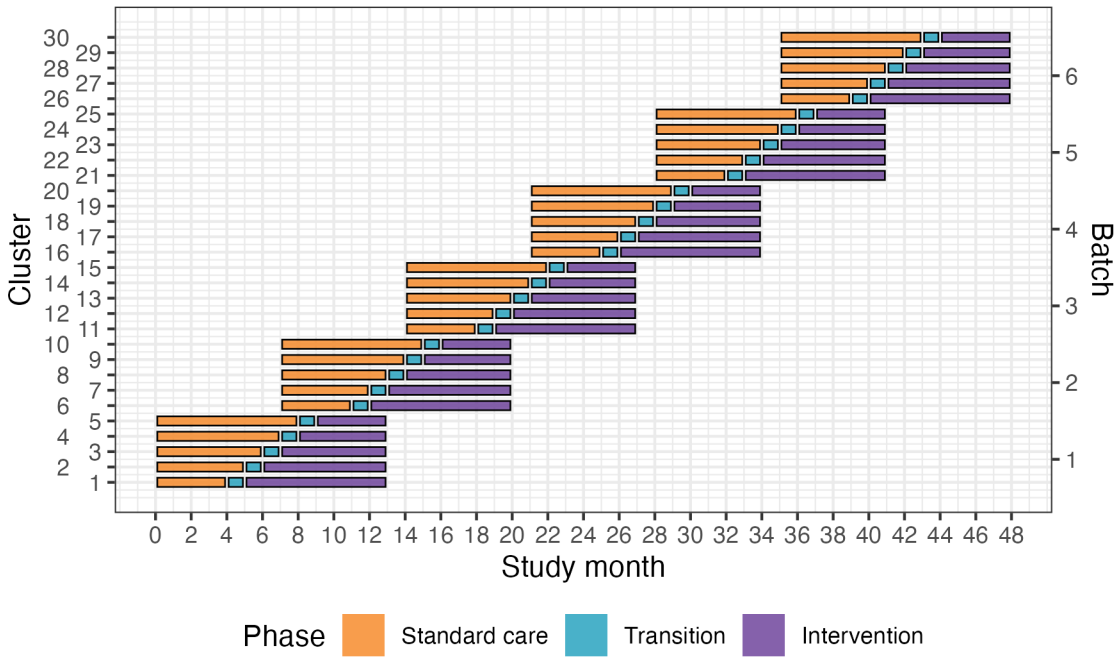


Figure 3: Trial design. Lines represent the duration of patient inclusion across clusters and phases. Clusters will be sequentially allocated to a batch based on when they enter the study. Within each batch clusters will then be randomised to an intervention implementation sequence.

8.3 Eligibility criteria

Our trial has eligibility criteria on the cluster and patient participant level.

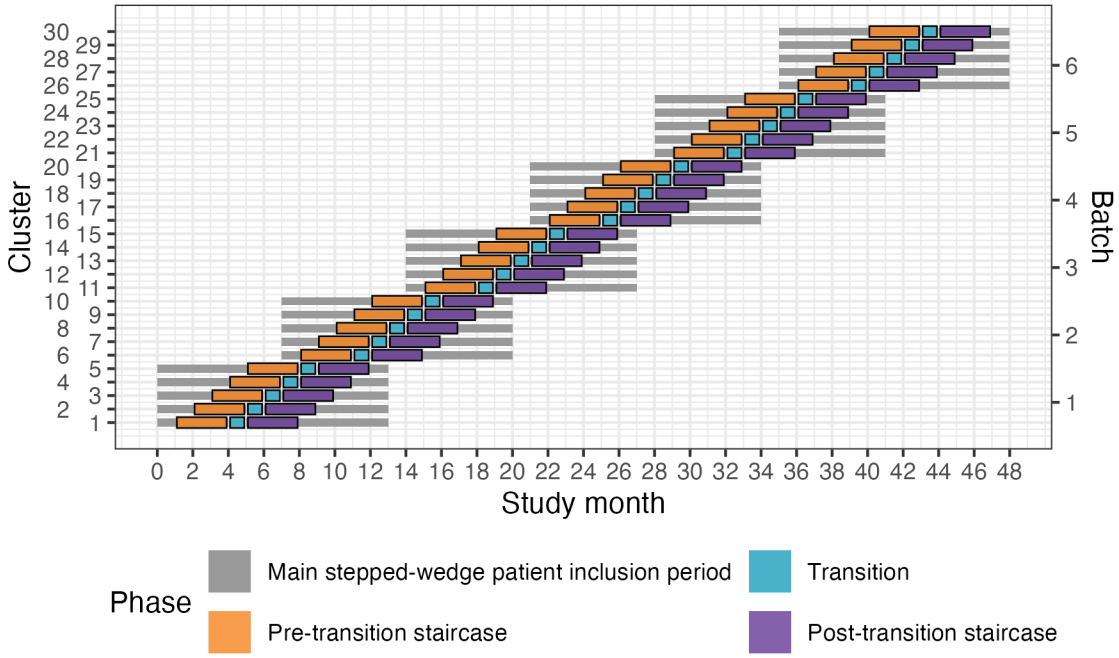


Figure 4: Nested staircase design. The design includes a random subset of patients presenting during the three months preceding the transition phase, and the three months following the transition phase.

8.3.1 Cluster selection

Clusters will be secondary or tertiary hospitals providing trauma care in India. This will be the unit of randomisation. Within each cluster, one or more units of physicians providing initial trauma care in the emergency department will be trained. These units already exist in the hospitals and rotate through the emergency department on specific days of the week.

8.3.1.1 Inclusion criteria

Clusters must meet the following criteria:

- admit or refer/transfer for admission at least 400 patients with trauma per year or 35 patients with trauma per month for at least the last six months;
- provide surgical and orthopaedic emergency services around the clock; and
- have at most 25% of physicians providing initial trauma care trained in a formalised trauma life support training programme, like ATLS® or Primary Trauma Care (PTC).

Additionally, the units to be trained within the cluster must meet the following criteria:

- admit or refer/transfer for admission at least 12 patients with trauma per month for at least the last six months; and
- no more than 25% of physicians providing initial trauma care trained in a formalised trauma life support training programme.

8.3.1.2 Exclusion criteria

Hospitals are excluded if they meet any of the following criteria:

- implement a formalised trauma life support training programme [1] during the trial period; or
- plan to implement or implement other major interventions[2] that affect trauma care during the trial period.

8.3.1.3 Screening

The trial management group will compile a list of hospitals with potentially eligible clusters and reach out to them to assess their interest in participating in the trial. We will then screen hospitals for eligibility based on the criteria above, using a two-step procedure. First, we will approach hospitals to complete an initial hospital screening instrument (see Appendix Initial hospital screening instrument (Section 20.1)). We will then discuss each eligible hospital individually in the Trial Management Group before deciding whether to include it in the trial. We have this discussion because we strive to include hospitals that to a large extent conducts primary resuscitation of trauma patients, rather than hospitals that primarily receives transferred patients from other hospitals, but this is difficult to formalise in the eligibility

[1] These include but are not limited to the National Emergency Life Support (NELS) programme, the Basic Trauma Life Support (BTLS) programme, the Pre-Hospital Trauma Life Support (PHTLS) programme, the Trauma Nursing Core Course (TNCC) and the Advanced Trauma Care for Nurses (ATCN) programme.

[2] These include but are not limited to implementing a trauma team approach, opening a trauma centre and implementing a trauma quality improvement programme.

criteria. We will then perform a more in-depth interview with selected hospitals (See Appendix In-depth hospital screening interview instrument (Section 20.2)). To avoid excluding hospitals we will also discuss plans to implement other potentially competing interventions during the trial period, and take these plans into account when assigning clusters to batches. For example, we are aware of the ongoing implementation of the National Emergency Life Support (NELS) programme in India, and will therefore not include hospitals that plan to implement this programme during the trial period. All screening steps and decisions will be logged using REDCap[38,39].

8.3.2 Patient participant selection

Patient participants are adult trauma patients who presents to the emergency department of participating hospitals and are admitted or transferred for admission.

8.3.2.1 Inclusion criteria

Patients participants must meet the following criteria:

- age of at least 15 years;
- trauma occurred less than 48 hours before arrival at the hospital;
- present to the emergency department of participating hospitals, with a history of trauma defined as having any of the reasons listed in the International Classification of Diseases chapter XX as the reason for presenting;
- admitted, or died between arrival at the hospital and admission, or referred/transferred from the emergency department of a participating hospital to another hospital for admission; and
- managed by a participating unit in the emergency department.

8.3.2.2 Exclusion criteria

Patients participants are excluded if they meet the following criteria:

- present with isolated closed extremity fracture; or
- are directly admitted to a ward without being seen by a physician in the emergency department.

8.3.2.3 Screening

Clinical research coordinators will screen patient participants either as they arrive to the emergency department or using emergency department registers. The patients or their representatives will receive written information about the study before they are discharged, including about their right to opt out at any time before final analysis. Phone numbers for out of hospital follow up will be extracted from the emergency department registers, and will be securely held only by the clinical research coordinators at each sites.

8.3.2.4 Withdrawal criteria

Patient participants can choose to withdraw their consent for collection of non-routinely recorded data at any time before the final analysis. If they withdraw their consent for this data collection the clinical research coordinator will not collect any more of this data, which also means that no further follow-ups will be conducted. They can also choose to have the data already collected about them removed from the trial at any time before final analysis of the data. Withdrawal of consent or removal of data from the trial will not affect their care in any way. If the patient participant withdraws consent, follow-up of this participant will be performed according to the participating hospitals routine.

8.4 Procedures

Table 3 shows an overview of trial procedures before and during patient admission, and Table 4 shows an overview of trial follow-up procedures. Clinical research coordinators will follow up patients daily until discharge to capture injury information. They will also follow up patients at 24 hours, 30 days and 90 days after arrival to the emergency department to capture mortality outcomes, and at 30 days and 90 days after arrival to the emergency department to capture functional outcomes and return to work. If patient participants are discharged before any of these follow-up time points, clinical research coordinators will follow up patients by phone.

8.5 Biological sampling procedures

This trial does not include biological sampling.

8.6 End of trial

The trial ends when the last patient participant has completed the last follow-up. The trial may be prematurely terminated if it this is necessary for safety reasons affecting the risk-benefit balance or if the recruitment of subjects cannot be met within reasonable time limits. If the trial is prematurely terminated or suspended, the investigator should immediately inform the subjects about this and ensure appropriate treatment and follow-up. Decisions on premature termination are taken by the joint Trial Steering and Data Monitoring Committee and Trial Management Group.

8.7 Intervention and control treatment

The intervention will be ATLS® training. The control will be standard care, meaning no formal trauma life support training. We will train the physicians that initially resuscitate and provide trauma care during the first hour after patient arrival at the emergency department. These physicians can be casualty medical officers, surgical residents, or emergency medicine residents, depending on the setup at each participating centre. The training will occur during the transition phase in each cluster. Our experience from our pilot study is that study sites adhere to the training slot allotted to them through the trial, so we judge the risk of clusters implementing ATLS® before their randomised implementation sequence as very low.

Table 3: Overview of trial procedures before and during patient admission

Procedure	Screening	Consenting	Initial assessment	In-hospital care
Eligibility criteria	√			
Study information[1]		√		
Informed consent[1]		√		
Baseline data collection			√	
Prehospital data collection			√	
ATLS adherence[2]			√	
ED data collection[3]			√	
Hospital data collection				√
Surgery data collection				√
Imaging data collection				√
Transfusion data collection				√
Injury data collection				√
Mortality data collection				√
Assessment of safety events				√

[1]Clinical research coordinators will inform patient participants about the study, including that they are free to withdraw their data from the study at any time, and approach them for informed consent for collection of non-routinely recorded data in person or telephonically.
[2]ATLS adherence will be assessed by observing the care provided to a random sample of patient participants.
[3]Emergency Department

Table 4: Overview of trial follow-up procedures

Procedure	Within 7 days of discharge	30 days	90 days
Mortality data collection[1]	√	√	√
EQ5D5L/WHODAS	√	√	√
Return to work		√	√
End of study			√

[1]Will be ascertained daily from when the patient participant arrive to hospital until they leave the hospital, are discharged or die.

We will train the number of units of physicians needed reach the required patient sample size, but estimate that this will require training an average of ten physicians per hospital, which should mean that we can train one to two units per hospital. This is possible because many hospitals in India organise physicians staffing their emergency departments in units, and the physicians in the same unit work together in the emergency department on the same days of the week. We will therefore collect data only on the days when these units work. The units selected to constitute a cluster from each hospital will be a convenience sample out of all eligible units in those hospitals.

Advanced Trauma Life Support® (ATLS®)[12] is a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. The programme was developed by the Committee of Trauma of the American College of Surgeons. The course includes initial treatment and resuscitation, triage and interfacility transfers. Learning is based on practical scenario-driven skill stations and lectures, and the course includes a final performance proficiency evaluation. Physicians will be trained in an accredited ATLS® training facility in India. We will assess adherence to ATLS principles before and after implementing ATLS training.

Standard care varies across hospitals in India, but trauma patients are initially managed by casualty medical officers, surgical residents, or emergency medicine residents. They are mainly first- or second-year residents who resuscitate patients, perform interventions and refer patients for imaging or other investigations. Compared with other settings where a trauma team approach is adopted, nurses and other healthcare professionals are only involved to a limited extent during the initial management.

8.7.1 Description of investigational medicinal products

This trial does not include any investigational medicinal products.

8.7.2 Auxiliary medicinal products

This trial does not include any auxiliary medicinal products.

8.7.3 Concomitant use of other medications or treatments

Other than implementing another formalised trauma life support training programme or other major interventions to change the care of trauma patients as specified in the exclusion criteria, concomitant use of other medications and treatments may be provided at the discretion of the investigators and will not be considered an exclusion criterion.

8.8 Randomisation

We will assign clusters to batches as they are found to be eligible and receive ethical approval. Batches will include clusters from hospitals in different regions to optimize trial logistics. We will randomise the clusters alloted to each batch to the different intervention implementation sequences within that batch. We will balance the randomisation within each batch on cluster size, defined as monthly volume of eligible patient participants, using covariate constrained randomisation. The cluster sizes are expected to vary between 12 and 20 patients per month, based on our previous experiences. We will conceal the

randomisation order for as long as it is logistically possible, considering that arrangements for sending physicians to ATLS® training need to be made in advance.

8.9 Blinding

It is not possible to blind a stepped-wedge trial, because all clusters receive the intervention.

8.10 Treatment after trial end

When the trial ends, the intervention will have been implemented in all clusters.

8.11 Outcomes

8.11.1 Primary outcome

The primary outcome will be in-hospital mortality within 30 days of arrival at the emergency department. We define this outcome as any death occurring in the hospital during the initial admission, up to 30 days post arrival at the emergency department. Clinical research coordinators will extract information on death from patient hospital records. If the patient has been transferred to another hospital, the clinical research coordinators will collect data on this outcome by calling the patient or a patient representative, or by contacting the hospital to which the patient was transferred. Data on this outcome will be collected continuously during the trial.

8.11.2 Secondary outcomes

8.11.2.1 Collected using the main stepped-wedge design

- All cause mortality within 24 hours, 30 days and three months of arrival at the emergency department. In contrast to the primary outcome, all-cause mortality includes all deaths regardless of setting. Data on this outcome will be collected in the same way as for the primary outcome.
- Length of emergency department stay. Data on this outcome will be collected from patient hospital records.
- Length of hospital stay. Data on this outcome will be collected from patient hospital records.
- Intensive care unit admission. Data on this outcome will be collected from patient hospital records.
- Length of intensive care unit stay. Data on this outcome will be collected from patient hospital records.
- Return to work at 30 days and three months after arrival at the emergency department. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged.

8.11.2.2 Collected using the nested staircase design

- Adherence to ATLS® principles during initial patient resuscitation, up to one hour after the physician has first seen the patient. This assessment will be done using a 14 item checklist covering the key steps of the ATLS® primary survey, which was modelled based on previous work on ATLS® adherence[40]. We will consider completion of all 14 steps as 100% adherence. The clinical research coordinators collecting the data will be trained by the trial team to do this, prior to the start of the trial. We will collect this data by observing the care of a random sample of patients. The sampling will be designed as a nested staircase design.
- Quality of life within seven days of discharge, and at 30 days and three months of arrival at the emergency department, measured by the official and validated translations of the EQ5D5L. This outcome is only relevant to alive patients. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged. We will collect this data using a nested staircase design.
- Disability within seven days of discharge, and at 30 days and three months of arrival at the emergency department, assessed using the WHO Disability Assessment Schedule 2.0 (WHODAS 2.0). This outcome is only relevant to alive patients. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged. This data will also be collected using a nested staircase design.

8.12 Handling of adverse and safety events

8.12.1 Definitions

8.12.1.1 Adverse event

Any untoward medical occurrence in a clinical trial subject and, which does not necessarily have a causal relationship with the treatment, can be an unfavorable and unintended sign (including an abnormal laboratory discovery), symptom or disease temporally associated with the inclusion in the trial, whether or not related to the trial.

8.12.1.2 Serious adverse event

Any untoward medical occurrence in a trial participant that:

- leads to death
- is life-threatening
- requires inpatient hospitalization or prolongation of existing hospitalization
- results in persistent or significant disability or incapacity
- results in a congenital anomaly/malformation

8.12.1.3 Safety event

Any unexpected serious complication that might occur as a consequence of the trial and that are not part of the natural history of trauma.

8.12.2 Reporting and assessment of adverse and safety events

In alignment with other current trials including critically ill patients[41], we will not collect adverse events or serious adverse events, because many of these events are expected in this patient population and we already collect many of these events, for example mortality, as part of our outcomes.

We will only report safety events, if they are life-threatening, prolong hospitalisation or result in meaningful harm to the participant. We cannot pre-define a comprehensive list of events that can be considered safety events, but will actively assess the presence of the following safety events:

- Prolonged mechanical ventilation (> 7 days)
- Initiation of renal replacement therapy
- Prolonged (> 2 days) or renewed (restart after at least 2 days without) use of vasopressors such as norepinephrine or vasopressin

These events are considered safety events because they suggest pulmonary, renal, septic or bleeding complications and an increase in their occurrence following ATLS® training could indicate that the intervention is harmful. These events therefore need to be tracked during the standard care phase as well as the intervention phase, but will only be considered indicative of harm related to the intervention if they occur more often during the intervention phase than during the standard care phase.

We will also report any other safety events that we identify during the trial, and the reporting of such will have to be based on the intuition of the clinical research coordinators and local investigators. Examples of such safety events could include missed injuries or missed investigations, which could be suspected if certain injuries or investigations were identified or conducted more often during the standard care phase than during the intervention phase.

All safety events will be recorded in the Case Record Form (CRF) and reported to the trial management team within 24 hours of its occurrence. The trial management team will then assess if the event can be considered related to the trial or the intervention within 24 hours of it being reported. Events that are considered probably related will be reported immediately to the joint Trial Steering and Data Monitoring Committee.

8.12.3 Follow up of safety events

All safety events should be followed up by the local investigator until they are fully evaluated.

8.13 Statistics

8.13.1 General principles

The primary analysis set will include all observations within clusters with available data and irrespective of the receipt of the intervention. Clusters and observations within clusters will be considered exposed to the intervention after the date at which the cluster was scheduled to transition. All data will be included with the exception of the transition phases. We will not adjust for multiplicity of analyses because none of the secondary outcomes will be singularly more important. However, all secondary

outcomes will be interpreted with due consideration for how all are affected by the intervention without putting any undue emphasis on a single outcome that might be statistically significant but where all others appear to have remained unchanged.

We will use a two-sided significance level of 5% and estimate 95% confidence intervals. The primary subgroup analyses will be based on geographical region because demonstrating the consistency of any effect across multiple regions will enhance the generalisibility of the results[3]. Additional subgroup analyses will include age across the groups older adolescents (15-19 years), young adults (20-24 years), adults (25-59 years), and older adults (60 years and older)[42]; sex; and the clinical cohorts blunt multisytem trauma, penetrating trauma, and severe isolated traumatic brain injury.

8.13.2 Analysis models

There are a number of requirements for the analysis model. Firstly, all analysis will consider the clustered nature of the design. Secondly, as the trial has only 30 clusters, it will be essential that the model allows for a correction due to the small number of clusters. Thirdly, as the design is a stepped-wedge study, we will adjust for temporal confounding using categorical effects for period of the study (month). Full details on how each of these will be undertaken, with justification is provided below[43].

For binary outcomes, a mixed effects binomial regression with a logit link will be used to estimate the odds ratio; and a binomial model with identity link used to estimate the risk difference. These models will be fitted using maximum likelihood, with a Laplace approximation to integrate over the random effects. If the binomial model with the identity link does not converge then only a odds ratio will be reported.

We will include fixed effects for period and a fixed effect for intervention exposure. The primary analysis will allow for clustering by as a random cluster and random cluster by period effect. To correct the potential inflation of the type I error rate due to small number of clusters, a correction for a small number of clusters will be applied, but the correction that will be selected will be based on the best available evidence available closer to the time, and it may differ for the outcomes collected via the complete and incomplete designs. In a sensitivity analysis we will explore if models with more complicated correlation structures are a better fit to the data. These models are not being used as our primary analysis models as there is limited understanding as to when such models will converge and how to choose between the various different correlation structures which might be plausible.

To this end we will additionally fit generalised linear mixed models (with same link functions and fixed effects as described above) to include a discrete time decay correlation structure including a random cluster effect with auto-regressive structure (AR(1)). To allow for the randomisation by batches, a different secular trend will be included for each batch (interaction between batch and period). For continuous, count and prevalence outcomes similar model-based approaches will be used but with appropriate links and distribution functions, using transformations where appropriate.

[3] **Note:** Batches will not be based on regions because it will be logistically more feasible to include clusters from different regions in each batch.

8.13.3 Additional sensitivity analyses

To additionally explore if the fixed period effect is both parsimonious and adequate to represent the extent of any underlying secular trend, we will model the time effect using a spline function. Models will also be extended to include random cluster by intervention effects (with a non-zero covariance term) to examine if results are sensitive to the assumption of no intervention by cluster interaction. Models will also be extended to include an interaction between treatment and number of periods since first treated, to examine if there is any indication of a relationship between duration of exposure to the intervention and outcomes.

This will allow us to different lag effects (whereby it takes time for the intervention to become embedded within the culture before its impact can properly start to be realised); as well as weaning effects (whereby the effect of the intervention starts to decrease – or fade). This type of analysis attempts to disentangle how some clusters end up having a long exposure to the intervention and others have a much shorter exposure time. A fully adjusted covariate analysis will additionally adjust for a set of pre-specified individual-level covariates of known prognostic importance.

8.13.4 Estimation and reporting of within cluster correlations

We will report time adjusted within-cluster correlations for all outcomes with 95% confidence intervals. We will report correlations from the different assumed correlation structures (so we will report intra-cluster correlations (ICC); within and between-period correlations; and within-period correlations and exponential decay). As well as reporting correlations we will additionally report all variance components. For all outcomes we will report correlations on the latent scale (i.e. proportions scale for binary outcomes) as is appropriate to inform future sample size calculations.

8.13.5 Sample size calculations

8.13.5.1 Main stepped wedge design and primary outcome

With 30 clusters across 6 batches and a total sample size of 4320 our study has ~90% power across different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC) to detect a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS® training (see Figure 5)[44]. This 5% absolute reduction in mortality is a clinically meaningful and conservative estimate based on our updated systematic review and pilot study[18,32]. We allowed for the clustered design, incorporated the one month transition period, and assumed: a discrete time decay correlation structure, an ICC of 0.02 (but considered sensitivity across the range 0.01-0.05), and a CAC of 0.9 (but considered sensitivity across the range 0.8-1.0), based on our pilot study and current guidance[45–48]. We included the CAC to allow for variation in clustering over time. We assume that each cluster will contribute approximately 12 observations per month to the analysis, but allowed for substantial variation in cluster sizes, based on our previous work.

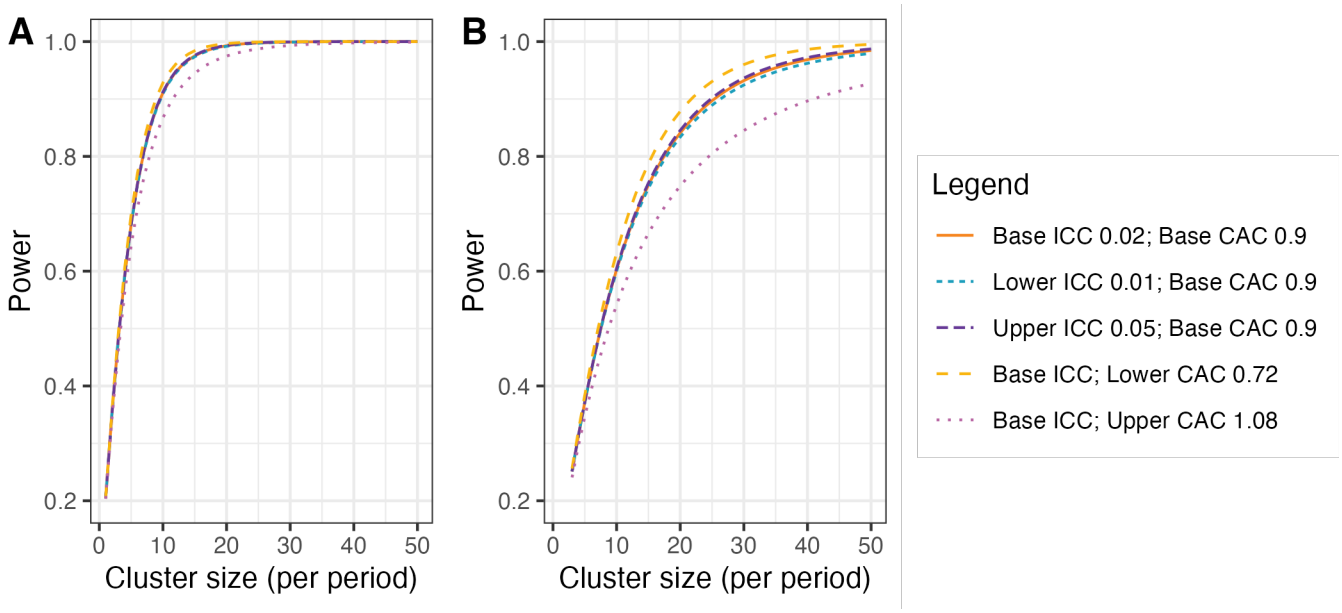


Figure 5: Power curves for different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC). **A)** Shows power curves assuming a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS® training. **B)** Shows power curves assuming a reduction in the primary outcome from 10% under standard care to 7.5% after ATLS® training. Under this scenario, we would need to increase the sample size per month to around 30 observations to achieve 90% powere under most combinations of CAC and ICC.

8.13.5.2 Nested staircase design and secondary outcomes

The secondary outcomes that will be measured using the nested staircase design are adherence to ATLS® principles during initial patient resuscitation, quality of life, and disability. The expected effect of the intervention on each of these outcomes are an improvement in adherence from 50% during standard care to 70% after training[49], an increase in EQ5D5L health status from 70 during care to 75 after training[50], and finally a decrease in disability from a baseline value of 25 during standard care to 22.5 after training[51]. For quality of life and disability, these effects correspond to standardised effect sizes, expressed as Cohen's d, of 0.5. With 30 clusters in total, six per sequence, for a discrete time decay correlation structure, with ICCs of 0.01 to 0.15 and a CAC of 0.8, there is >80% power to detect these effects by including four patients in each cluster in each period. Accounting for loss to follow-up, we will include at least six patients per cluster per month. These patients will be a random subset of patients included in the primary outcome data collection during the staircase months. The random subset will be selected using simple random sampling on the shift level, meaning that the timing of the clinical research coordinator's shift will be randomised to cover approximately eight hours during the morning, afternoon, or night shift. Adherence to ATLS® principles, quality of life, and disability will be measured in all patients included during these shifts. In each hospital, the number of shifts that will be randomised will be determined by the volume of patients included during the months preceding the staircase months.

8.13.6 Interim analysis

There will be one interim analysis after half of the batches have completed the trial. The interim analysis will be assessed by the joint Trial Steering and Data Monitoring Committee. The purposes of this interim analysis will be to:

- assess the trial's feasibility and recommend stopping the trial if the trial is not feasible, for example if hospitals fail to adhere to the randomisation schedule or if there are substantial missing data in outcomes;
- compare characteristics across intervention conditions to monitor for differential recruitment/ascertainment between intervention and control.

8.14 Quality control and quality assurance

The George Institute for Global Health - India will ensure proper conduct of the trial through quality control measures including on-site training of personnel, standard operating procedures, ongoing quality metrics assessment, review of missing data and outliers, and round-the-clock availability of coordinating center personnel and Principal Investigators. The trial will strictly follow ICH GCP principles, Indian regulations, and George Institute procedures. The trial operations staff from the George Institute India will train local investigators, and trial site staff, before the trial, with continuous documentation in the site master file. All documentation will be stored securely and retained according to regulatory requirements.

8.15 Quality assurance and oversight

The Trial Management Group and Trial Team, comprising key project leaders and managers, will play a pivotal role in ensuring the highest standards of quality assurance and effective sponsor oversight throughout the trial. These groups will be responsible for facilitating consistent communication, maintaining fidelity in study implementation, and overseeing the quality of data collection.

To achieve these objectives, the groups will implement a comprehensive communication plan and provide extensive training to site personnel. The training will cover not only the study protocol but also practical aspects of various systems, supplemented by both written and electronic materials designed to educate study and clinical emergency staff.

The trial's quality assurance systems will be meticulously designed based on a thorough risk analysis. A key component of our quality assurance strategy will include the development and implementation of detailed operational manuals and regular meetings. These tools and interactions will ensure that all trial personnel will be used to uphold the trial's quality standards.

Central to our oversight approach will be a comprehensive monitoring and auditing plan. This plan will be tailored based on the identified risks associated with the trial. Through these comprehensive measures, the trial management group, in conjunction with the hospital staff, will ensure that the trial is conducted with the utmost rigor, adhering to the highest standards of quality assurance and effective sponsor oversight.

8.16 Monitoring

We will implement a multi-tiered monitoring strategy, including centralized data consistency checks, statistical monitoring, and selective on-site evaluations. Key integrity measures include source data verification, data entry validation, and regular audits. Any protocol deviations will be thoroughly documented, with serious breaches promptly addressed to ensure data integrity. Monitors from coordinating centres will assist investigators in maintaining high ethical, scientific, technical, and regulatory quality. Monitoring visits will review protocol adherence, participant recruitment, adverse event reporting, compliance with study procedures, and regulatory adherence. Regular remote monitoring of the web-based database will be conducted to ensure data integrity, using validation and consistency rules and regular data cleaning. The Trial Team and Trial Management Group will monitor baseline characteristics, opt-in consent rates and differential opt-in consent rates across trial arms, follow-up rates, CRF return and completeness rates, and safety data.

9 Deviations, serious breaches and other reporting obligations

The responsible investigator shall, without delay, report to the sponsor any serious breaches and deviations from the trial protocol, ICH-GCP and other regulations that significantly and directly affect, or with high likelihood could affect, the subjects' safety and integrity or the reliability and robustness of the data generated in the trial. The sponsor should assess the suspected serious breach and the consequences of deviations that have occurred. Minor deviations that do not affect subjects' integrity or safety, nor

significantly affect the trial's scientific value, are documented in the trial documentation of the principal investigator and the sponsor and appropriate measures shall be taken. The deviations must be recorded in the clinical trial report.

10 Audits and inspections

Authorized representatives for the sponsor and Competent Authorities (CA) may carry out audits or inspections at the trial site, including source data verification. The investigator must ensure that all source documents are available for audits and inspections. The purpose of an audit or inspection is to systematically and independently review all trial-related activities and documents, to determine whether these activities were performed, registered, analyzed and reported correctly according to protocol, ICH-GCP and applicable regulations.

11 Ethics

11.1 Compliance to the protocol, ICH-GCP and regulations

The trial will be performed in compliance with this clinical trial protocol, the Declaration of Helsinki, ICH-GCP (Good Clinical Practice), and current national regulations governing this clinical trial. This is to ensure the safety and integrity of the trial subjects as well as the quality of the data collected.

11.2 Ethical review of the trial

The final protocol will be submitted for ethical review at all participating hospitals, where possible, as well as the The George Institute for Global Health in India and Swedish Ethical Review Atuhortiy.

11.3 Procedure for obtaining consent

In this trial, consent refers to data collection, as patients cannot opt out of the intervention. This is because the intervention is implemented at the cluster level, involving training physicians in ATLS®. It is unreasonable to expect these physicians to temporarily disregard their training. Patient participants will be included in this trial under the following modes of consent:

- Opt out consent for **routinely recorded data and measurement of adherence to ATLS® principles**. Consent for the collection of routinely recorded data, either through interviews or by extracting information from medical records, as well as for the measurement of adherence to ATLS® principles, will be presumed unless explicitly declined. This approach is justified because the trial is considered to pose minimal risk and because data collection will be non-invasive. Additionally, obtaining consent specifically for the measurement of adherence to ATLS® principles could interfere with the provision of care and cause undue stress for the patient and their representatives. Patients, or their legally authorized representatives, will be provided

with written information about the study upon their arrival at the hospital. The variables assumed to be routinely recorded are listed in Variables (Section 18.2).

- Opt in consent and assent for **non-routinely recorded data**. Informed consent for non-routinely recorded data will be actively sought from patient participants or their legally authorized representative. For participants who are between 15 and 18 years of age we will obtain both the assent of the participant as well as the consent of their guardian or legally authorized representative. The clinical research coordinators will approach patient participants and their representatives after admission. The consent and assent will be written for patient participants who are admitted to the hospital and verbal for participants who are transferred or discharged before the clinical research coordinators have had an opportunity to approach them. The verbal consent will be audio recorded.
- Waiver of informed consent for patients who are unconscious or otherwise unable to provide consent and do not have a legally authorized representative. This group represents the most severly injured patients and they have to be included to make the trial representative of the entire population of trauma patients. Patients participants who regain consciousness will be informed about the study and asked for consent for collection of non-routinely recorded data.

11.4 Data protection

All data will be handled according to the Indian Council of Medical Research's guidelines and standard operating procedures of the George Institute for Global Health India on data security and protection. Trial data will be shared via the trial electronic CRF (eCRF) throughout the trial. The eCRF will be accessible via VPN with a two-factor authentication and the data will be held on a secure server. All investigators and trial site staff involved in this trial must comply with the requirements of the ICMR Guidelines on data security and protection. The participant information sheet provided to participants, will inform them how:

- the trial data will be collected, used and disclosed;
- how trial data are stored to maintain confidentiality in accordance with national data legislation; and
- for verification of the data, representatives delegated by the sponsor, as well as relevant authorities, may require access to parts of medical records or trial records that are relevant to the trial, including the patient participant's medical history.

12 Insurances

The George Institute for Global Health, India is responsible for ensuring that any insurance cover required to cover the set-up, management and conduct of the study in India has been obtained. The George Institute for Global Health, India is also responsible for ensuring that India Sites have been obtained and/or will obtain insurance prior to the opening of the study in India and shall be maintained for the duration of the study and for an appropriate period thereafter. This includes being responsible for ensuring that there is appropriate insurance for the duration of the study to cover against claims for compensation by participants arising out of their participation in the trial in India. Compensation in

case of injury or death will be provided by the George Institute for Global Health, India according to the regulations outlined in rules 39, 40 and 42 of the New Drugs and Clinical Rules (2019). x

13 Substantial changes to the trial

Substantial changes to the signed clinical trial protocol are only possible through approved protocol amendments and by agreement between the sponsor and the principal investigator.

14 Trial organisation

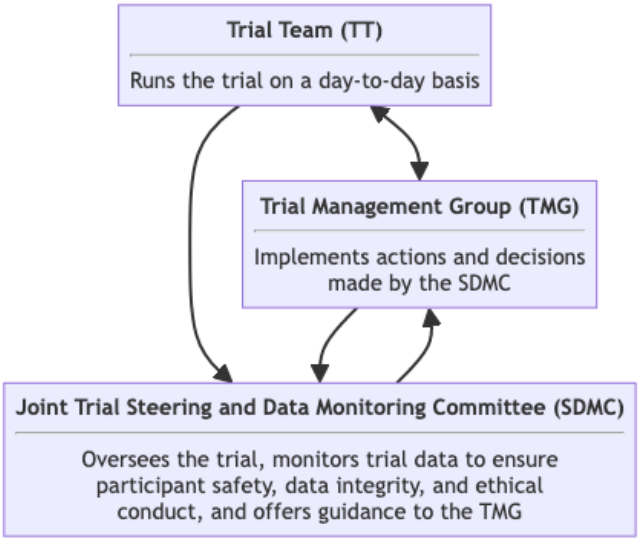


Figure 6: Trial organisation overview.

Trial management and oversight is governed by three trial committees and groups: the Trial Team (TT), the Trial Management Group (TMG), the joint Trial Steering and Data Monitoring Committee (SDMC). These groups and their relationships are briefly described in Figure 6. Details about each committee and group are available in their respective charter.

14.1 Trial team

Responsibility

To run the trial on a day-to-day basis, maintain trial databases, randomise clusters, ensuring complete and correct data, preparing reports for meetings (including those of the TMG and SDMC) and dealing with research governance and, if appropriate, regulatory matters.

Composition

Includes the project manager, clinical research associates, principal investigator and co-investigators as needed.

Relationships

Reports to the TMG and SDMC. Operationalises decisions made by the TMG.

Meeting frequencies

As often as needed, often weekly or bi-weekly.

14.2 Trial Management Group (TMG)

Responsibility

To manage the trial, including its clinical and practical aspects.

Composition

Includes members with broad expertise appropriate to the trial. The TMG will be chaired by the Principal Investigator.

Relationships

Receives reports from TT. Provides input to the SDMC. Implements decisions made by the SDMC.

Meeting frequencies

Monthly to every six months.

14.3 Joint Trial Steering and Data Monitoring Committee (SDMC)

Responsibility

The SDMC's responsibility is to oversee the trial, review results of interim analyses and safety events reported by the TMG, and review trial data for each batch, assessing data quality, completeness, cluster performance in recruitment and loss to follow-up rates, and external factors affecting trial validity, safety, or ethics. This committee also offer guidance to the TMG.

Composition

A majority of independent members, including a chair and three additional external experts specializing in the clinical area, biostatistics, and a community or patient representative, as well as and a minority of members with a direct interest in the trial, including the principal investigator. The chair should be independent of the trial, and the coordinating institutions Karolinska Institutet and The George Institute for Global Health.

Relationships

Receives reports from the trial team and TMG.

Meeting frequencies

After the completion of each batch, but may be more frequent if needed.

15 Funding

- Swedish Research Council (reg. no. 2023-03128)
- Laerdal Foundation (reg. no. 2023-0297)

16 Special considerations

16.1 Funding

This trial is not yet fully funded. The Trial Management Group has decided to proceed with the trial with the expectation that additional funding will be secured. The Trial Steering Committee will be informed of the funding status at each meeting. If funding is not secured, the trial will be stopped. This will likely result in an underpowered trial. The justification for this decision is that the intervention is considered standard of care in many countries and the data collection is considered minimal risk. There is therefore a very small risk of harm to patient participants, but a potential direct benefit to those patient participants who receive the intervention. The benefit-risk ratio is therefore considered to be favourable, even in the case of an underpowered trial.

17 Notification of trial completion, reporting, and publication

The trial will be reported to the Funders within a year of completion. The results of the trial will also be prepared as manuscripts for publication. Authorship on trial manuscripts will be based on the International Committee of Medical Journal Editors (ICMJE) criteria[52]:

- Substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work; AND
- Drafting the work or reviewing it critically for important intellectual content; AND
- Final approval of the version to be published; AND
- Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

In addition to being accountable for the parts of the work done, an author should be able to identify which co-authors are responsible for specific other parts of the work. In addition, authors should have confidence in the integrity of the contributions of their co-authors.

The most recent version of the ICMJE criteria will be adhered to. We will also use the ICMJE criteria for non-author contributorship.

Before work on a trial manuscript is initiated, a writing group will be formed and first and last authors will be designated. This writing group will be formed by discussion in the Trial Management Group.

18 Collection, handling, and archiving of data

Clinical research coordinators will collect data using a paper based CRF, see Appendix Case Record Form (Section 20.3), which is then transferred to an eCRF. All trial data in the CRF must be extracted from and be consistent with the relevant source documents. The eCRF will be accessible to trial coordinators, data managers, the Investigators, Clinical Trial Monitors, Auditors, and Inspectors as required. All data will be registered, managed, and stored in a manner that enables correct reporting, interpretation, and verification. The complete Trial Master File, as well as source documents, will be archived for at least 10 years after the trial is completed. Source data in the medical records system are stored and archived in accordance with national regulations. Metadata will be publicly accessible via a persistent DOI, and anonymised data will be released upon project completion. A detailed data management plan is available here https://doi.org/10.5281/zenodo.7748764.

18.1 Source data

The source data for each variable is given in Variables (Section 18.2). Whenever medical records are the source data, this includes imaging and lab reports. Whenever an interview is given as the source, the CRF will constitute the source data, as this is where the responses to questions will be recorded. The local investigator must keep source documents for each patient participant in the trial. A document describing what has been classified as source data in the trial (source data reference document) will be included in the Investigator Site File (ISF). The investigator must ensure that all source documents are accessible for monitoring and other quality control activities. Source data is further defined before trial start at each individual site and can, in cases where source data is not registered in another document, consist of the CRF. This should be decided in consultation with the monitor and clearly stated in the source data reference document. Access to trial-related documentation, such as patient participants' medical records, CRFs, other source data and other trial documentation will be provided for monitoring and auditing purposes. Access will also be granted in the context of regulatory inspections.

18.2 Variables

18.2.1 Screening

- **Screening ID**
- **Date of screening**
- **Date of data entry**
- **Is the patient at least 15 years old?** Source: Medical record or interview

  - Yes
  - No

- **Did the patient present with a history of trauma defined as having any of the reasons listed in the International Classification of Diseases chapter XX as the reason for presenting? Please see https://icd.who.int/browse10/2019/en#/XX for a complete list of ICD-10 codes** Source: Medical record or interview

  - Yes
  - No

- **Did the trauma occur less than 48 hours before arrival to the hospital?** Source: Medical record or interview

  - Yes
  - No

- **Was the patient admitted?** Source: Medical record

  - Yes
  - No

- **Did the patient die after arrival but before admission?** Source: Medical record

  - Yes
  - No

- **Was the patient transferred to another hospital for admission?** Source: Medical record

  - Yes
  - No

- **Did the patient present with isolated closed extremity fracture?** Source: Medical record

  - Yes
  - No

- **Was the patient directly admitted to a ward without being seen by a physician in the emergency department?** Source: Medical record

  - Yes
  - No

- **Is the patient eligible? (Eligible=1; Not Eligible=0)**
- **Any other comments?**

18.2.2 Consent

- **Is this patient included under the waiver of informed consent because the patient is unconscious or otherwise unable to provide consent and do not have a legally acceptable representative?**
  - Yes
  - No
- **Did the participant/ or legally acceptable representative (LAR) provided consent for collection of non-routinely recorded data**
  - Yes
  - No
- **Who gave consent for collection of non-routinely recorded data?**
  - Patient participant
  - Legally acceptable representative
- **What is the LAR's relationship with the participant?**
- **Why was Legally acceptable representative (LAR) approached for consent for collection of non-routinely recorded data?**
  - The participant is incapacitated because of the trauma
  - The participant is younger than 18 years
- **Date when participant or legally acceptable representative (LAR) gave consent for collection of non-routinely recorded data.**
- **How did the participant or legally acceptable representative (LAR) consent for collection of non-routinely recorded data?**
  - In writing
  - Verbally
- **Reconsenting done?**
  - Yes
  - No
- **Date when the participant was reconsented?**
- **Why consenting was not done?**
- **Did the minor give assent for collection of non-routinely recorded data?**
  - Yes

  - No

- **Date when the minor gave assent for collection of non-routinely recorded data.**
- **In case the minor refused to participate, date when minor refused**
- **Is the participant or LAR wants to opt out from study?**
  - Yes
  - No
- **Who opted-out of the routinely recorded data (in-hospital)?**
  - Patient participant
  - Legally acceptable representative (LAR)
- **Date when participant or legally acceptable representative (LAR) opted-out.**
- **Did the participant or legally acceptable representative (LAR) suggested to delete all the previously recorded data?**
  - Yes
  - No
- **Any other comments?**

18.2.3 Consent withdrawn

- **Does the participant or legally acceptable representative (LAR) want to withdraw the consent?**
  - Yes
  - No
- **Date of consent withdrawal for follow-up data collection.**
- **Procedure(s) for which consent has been withdrawn**
  - Data collection prior to withdrawal
  - All data collection after withdrawal
  - Both
- **Any other comments?**

18.2.4 Baseline

- **Age in years** Source: Medical record of interview
- **Sex** Source: Medical record of interview
  - Female
  - Male
  - Other
  - Not known
- **Current marital status** Requires opt-in consent, not routinely recorded. Source: Interview
  - Never married
  - Currently married
  - Separated
  - Divorced
  - Widowed
  - Cohabiting
  - Not known
- **Education level** Requires opt-in consent, not routinely recorded. Source: Interview
  - Not attended school
  - Primary school
  - Secondary school
  - Higher secondary school
  - Graduate
  - Post graduate and above
  - Other
  - Not known
- **Main work status** Requires opt-in consent, not routinely recorded. Source: Interview
  - Paid work, such as daily wage earner, teacher, factory worker and government employee
  - Self-employed, such as own your business or farming
  - Non-paid work, such as volunteer or charity
  - Student
  - Keeping house/homemaker
  - Retired
  - Unemployed (health reasons)
  - Unemployed (other reasons)
  - Other
  - No income
  - Not known

- **Income level in INR per month** Requires opt-in consent, not routinely recorded. Source: Interview
  - Below 10,000
  - 10,001-20,000
  - 20,001-30,000
  - 30,001-50,000
  - 50,001-80,000
  - 80,001-1,00,000
  - Above 1,00,000
  - No income
  - Not known
- **Mechanism of injury description**
- **Mechanism of injury ICD 10 code** Coded using ICD 10. Please see https://icd.who.int/browse10/2019/en#/XX for a complete list of ICD-10 codes. Source: Medical record
  - 1,109 ICD-10 codes: V01–V99 (872), W00–W99 (88), X00–X99 (93), Y00–Y36 (56). The full list is available in the eCRF.
- **Clinical Frailty Scale** Source: Medical record or treating physician
  - Very Fit
  - Fit
  - Managing well
  - Living with very mild frailty
  - Living with mild frailty
  - Living with moderate frailty
  - Living with severe frailty
  - Living with very severe frailty
  - Terminally ill
  - Not known
- **Comorbidities (Charlson Comorbidity Index)** Source: Medical record, treating physician or interview
  - Myocardial infarction
  - Congestive heart failure
  - Peripheral vascular disease
  - Cerebrovascular disease
  - Dementia
  - Chronic pulmonary disease
  - Rheumatologic disease
  - Peptic ulcer disease
  - Liver disease

  - Diabetes
  - Hemiplegia or paraplegia
  - Renal disease
  - Malignancy
  - Leukemia
  - Lymphoma
  - AIDS
  - Not known
  - None
  - Other

- **Severity of liver disease** Source: Medical record, treating physician or interview

  - Mild
  - Moderate or severe
  - Not known

- **Severity of diabetes** Source: Medical record, treating physician or interview

  - Controlled
  - Uncontrolled
  - Not known

- **Severity of malignancy** Source: Medical record, treating physician or interview

  - Localized
  - Metastatic tumor
  - Not known

- **Any other comments?**

18.2.5 Prehospital

- **Date of injury** Source: Medical record of interview
- **Time of injury**
- **Mode of transport to the participating hospital** Source: Medical record of interview

  - Ambulance
  - Police
  - Private vehicle
  - Walking
  - Others
  - Not known

- **Referred or transferred to the participating hospital from another hospital** Source: Medical record of interview
  - Yes
  - No
  - Not known
- **Date of referring or transferring**
- **Time of referring or transferring**
- **Any other comments?**

18.2.6 ATLS adherence

- **Airway patency checked** Source: Observation
  - Yes
  - No
- **Chest wall palpated** Source: Observation
  - Yes
  - No
- **Breath sounds checked** Source: Observation
  - Yes
  - No
- **Respiratory rate measured** Source: Observation
  - Yes
  - No
- **Saturation (SpO2) measured** Source: Observation
  - Yes
  - No
- **Heart rate measured** Source: Observation
  - Yes
  - No
- **Blood pressure measured** Source: Observation
  - Yes
  - No
- **Abdomen palpated** Source: Observation

  - Yes
  - No

- **Thighs palpated** Source: Observation
  - Yes
  - No

- **IV access obtained** Source: Observation
  - Yes
  - No

- **GCS checked** Source: Observation
  - Yes
  - No

- **Pupils checked** Source: Observation
  - Yes
  - No

- **Patients exposed for assessment**
  - Yes
  - No

- **Temperature measured** Source: Observation
  - Yes
  - No

- **Which airway interventions were performed?** Source: Observation
  - None
  - Manual airway procedure such as chin lift or jaw thrust
  - Nasopharyngeal or Oropharyngeal airway inserted
  - Supraglottic airway device
  - Tracheal intubation
  - Surgical airway
  - Other
  - Not known

- **Were airway interventions performed while minimising c-spine movement?** Source: Observation
  - Yes
  - No
  - Not known

- **Which breathing interventions were performed?** Source: Observation
  - None
  - Oxygen applied
  - Intracostal drain placement
  - Other
  - Not known
- **Which circulation interventions and adjuncts were performed?** Source: Observation
  - None
  - Control of external bleeding
  - Fluid bolus
  - Blood transfusion
  - eFast
  - Pelvic binder applied
  - Reduction of highly displaced fracture
  - Other
  - Not known
- **Which disability intervention was performed?** Source: Observation
  - None
  - Placement of definitive airway if the patient had a GCS of 8 or less
  - Log Rolling
  - Spine board during transportation
  - Other
  - Not known
- **Which exposure intervention was performed?** Source: Observation
  - None
  - Covered with warmer or blanket
  - Warm fluids administered
  - Other
  - Not known
- **Any other comments?**

18.2.7 Emergency department

- **Date of arrival to the emergency department at the participating hospital.** Source: Medical record of interview
- **Time of arrival to the emergency department at the participating hospital**
- **First recorded systolic blood pressure (mmHg)** Source: Medical record
- **First recorded diastolic blood pressure (mmHg)** Source: Medical record

- **First recorded heart rate (beats per minute)** Source: Medical record
- **First recorded respiratory rate (breaths per minute)** Source: Medical record
- **First recorded Glasgow Coma Scale** Source: Medical record
- **First recorded body temperature (°F)** Source: Medical record
- **First recorded oxygen saturation (%)** Source: Medical record
- **Emergency department disposition** Source: Medical record
  - Admitted
  - Referred or transferred for admission
  - Dead
  - Others
  - Not known
- **Date of death**
- **Time of death**
- **Date and time of referral or transfer for admission** Source: Medical record
- **Any other comments?**

18.2.8 Hospital

- **Date of admission to the participating hospital** Source: Medical record
- **Time of admission to the participating hospital** Source: Medical record
- **Type of admitting ward** Source: Medical record
  - General surgery
  - Orthopaedics
  - Neurosurgery
  - Intensive care unit
  - High dependency unit
  - Medicine
  - Trauma ward
  - Not known
- **Ward name or number** Source: Medical record
- **Admitted to intensive care unit during admission** Source: Medical record
  - Yes
  - No
  - Not known

- **Date of first intensive care unit admission** Source: Medical record
- **Time of first intensive care unit admission** Source: Medical record
- **Was the participant discharged from intensive care unit?**
  - Yes
  - No
- **Date of first intensive care unit discharge** Source: Medical record
- **Time of first intensive care unit discharge** Source: Medical record
- **Hospital disposition** Source: Medical record
  - Alive
  - Dead
  - Transferred for admission
  - Not known
- **Was the patient transferred to another hospital for admission?** Source: Medical record
  - Yes
  - No
  - Not known
- **Date of discharge or transfer from participating hospital** Source: Medical record
- **Time of discharge or transfer from participating hospital** Source: Medical record
- **Any other comments?**

18.2.9 Surgery

- **Surgery done?**
  - Yes
  - No
- **Date of surgical procedure** A surgical procedure is defined as any procedure performed in the operating room, interventional dropdownlogy suite, or at the bedside, requiring general or regional anesthesia. Source: Medical record
- **Time of surgical procedure** A surgical procedure is defined as any procedure performed in the operating room, interventional dropdownlogy suite, or at the bedside, requiring general or regional anesthesia. Source: Medical record
- **Preoperative ASA score** Source: Medical record or treating physician
  - A normal healthy patient

  - A patient with mild systemic disease
  - A patient with severe systemic disease
  - A patient with severe systemic disease that is a constant threat to life
  - A moribund patient who is not expected to survive without the operation
  - A declared brain-dead patient whose organs are being removed for donor purposes
  - Not known

- **Description of procedure** Source: Medical record
- **Procedure coded according to SNOMED CT** Source: Medical record
- **Any other comments?**

18.2.10 Imaging

- **Date and time of imaging** Source: Medical record
- **Time of imaging** Source: Medical record
- **Type of imaging** Source: Medical record

  - Ultrasound
  - X-ray
  - Computed Tomography (CT)
  - Magnetic Resonance Imaging (MRI)

- **Any other comments?**

18.2.11 Transfusion

- **Transfusion done?**

  - Yes
  - No

- **Date of transfusion** Source: Medical record
- **Time of transfusion** Source: Medical record
- **Type of blood product** Source: Medical record

  - Packed red blood cells
  - Platelets
  - Fresh frozen plasma
  - Whole blood
  - Other

- **Number of units transfused** Source: Medical record
- **Any other comments?**

18.2.12 Injury

- **Injury description** Source: Medical record
- **Injury ICD 10 code** Coded using ICD 10. Source: Medical record
  - 995 ICD-10 codes: S00–S99 (694), T00–T74 (301). The full list is available in the eCRF.
- **Injury source data** Source: Medical record
  - Medical record
  - X-ray report
  - CT-report
  - Surgical notes
- **AIS code**
- **Any other comments?**

18.2.13 Individual mortality status

- **Is the patient dead?** Source: Medical record or interview
  - Yes
  - No
- **Date of death** Source: Medical record or interview
- **Time of death**
- **Any other comments?**

18.2.14 Quality of life (EQ5D5L)

- **Date of filling this form**
- **First, I would like to ask you about MOBILITY. Would you say that:** Requires opt-in consent, not routinely recorded. Source: Interview
  - You have no problems in walking about?
  - You have slight problems in walking about?
  - You have moderate problems in walking about?
  - You have severe problems in walking about?
  - You are unable to walk about?
- **Next, I would like to ask you about SELF-CARE. Would you say that:** Requires opt-in consent, not routinely recorded. Source: Interview
  - You have no problems washing or dressing yourself?
  - You have slight problems washing or dressing yourself?

  - You have moderate problems washing or dressing yourself?
  - You have severe problems washing or dressing yourself?
  - You are unable to wash or dress yourself?

- **Next, I would like to ask you about USUAL ACTIVITIES, for example, work, study, housework, family or leisure activities. Would you say that:** Requires opt-in consent, not routinely recorded. Source: Interview

  - You have no problems doing your usual activities?
  - You have slight problems doing your usual activities?
  - You have moderate problems doing your usual activities?
  - You have severe problems doing your usual activities?
  - You are unable to do your usual activities?

- **Next, I would like to ask you about PAIN OR DISCOMFORT. Would you say that:** Requires opt-in consent, not routinely recorded. Source: Interview

  - You have no pain or discomfort?
  - You have slight pain or discomfort?
  - You have moderate pain or discomfort?
  - You have severe pain or discomfort?
  - You have extreme pain or discomfort?

- **Finally, I would like to ask you about ANXIETY OR DEPRESSION. Would you say that:** Requires opt-in consent, not routinely recorded. Source: Interview

  - You are not anxious or depressed?
  - You are slightly anxious or depressed?
  - You are moderately anxious or depressed?
  - You are severely anxious or depressed?
  - You are extremely anxious or depressed?

- **I would now like you to tell me the point on this line where you would put your health TODAY.(Note to interviewer: mark the line at the point indicating the respondent's health today.)** Requires opt-in consent, not routinely recorded. Source: Interview
- **Any other comments?**

18.2.15 Disability (WHODAS 2.0)

- **Date of form filling**
- **Who are you interviewing?** Requires opt-in consent, not routinely recorded. Source: Interview

  - Patient participant
  - Patient representative

- **What is the relationship between the representative and the participant?** Requires opt-in consent, not routinely recorded. Source: Interview
  - Husband or wife
  - Parent
  - Son or daughter
  - Brother or sister
  - Other relative
  - Friend
  - Professional carer
  - Other (specify)
- **Standing for long periods such as 30 minutes?** Requires opt-in consent, not routinely recorded. Source: Interview
  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do
  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do
- **Taking care of your household responsibilities?** Requires opt-in consent, not routinely recorded. Source: Interview
  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do
- **Learning a new task, for example, learning how to get to a new place?** Requires opt-in consent, not routinely recorded. Source: Interview
  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do
  - None
  - Mild

  - Moderate
  - Severe
  - Extreme or cannot do

- **How much of a problem did you have joining in community activities (for example, festivities, religious or other activities) in the same way as anyone else can?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **How much have you been emotionally affected by your health problems?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **Concentrating on doing something for ten minutes?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do
  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **Walking a long distance such as a kilometre [or equivalent]?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do
  - None

  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **Washing your whole body?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **Getting dressed?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do
  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **Dealing with people you do not know?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **Maintaining a friendship?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do
  - None
  - Mild
  - Moderate
  - Severe

  - Extreme or cannot do

- **Your day-to-day work/school?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **Taking care of his or her household responsibilities?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **How much of a problem did he or she have joining in community activities (for example, festivities, religious or other activities) in the same way as anyone else can?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **How much has your relative been emotionally affected by his or her health condition?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **Washing his or her whole body?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe

  - Extreme or cannot do

- **Dealing with people he or she does not know?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **His or her day-to-day work/school?** Requires opt-in consent, not routinely recorded. Source: Interview

  - None
  - Mild
  - Moderate
  - Severe
  - Extreme or cannot do

- **Overall, in the past 30 days, how many days were these difficulties present?** Requires opt-in consent, not routinely recorded. Source: Interview
- **In the past 30 days, for how many days were you totally unable to carry out your usual activities or work because of any health condition?** Requires opt-in consent, not routinely recorded. Source: Interview
- **In the past 30 days, not counting the days that you were totally unable, for how many days did you cut back or reduce your usual activities or work because of any health condition?** Requires opt-in consent, not routinely recorded. Source: Interview
- **Any other comments?**

18.2.16 Return to work

- **Date of form filling**
- **Did the participant return to work?**

  - Yes
  - No

- **Date and time of return to work** Requires opt-in consent, not routinely recorded. Source: Interview
- **Work status** Requires opt-in consent, not routinely recorded. Source: Interview

  - Paid work
  - Self-employed, such as own your business or farming

  - Non-paid work, such as volunteer or charity
  - Student
  - Keeping house/homemaker
  - Not known

- **Any other comments?**

18.2.17 Safety events

- **Is any safety event reported?**

  - Yes
  - No

- **Date reported to trial management team of safety event**
- **Type of safety event** Source: Medical record or treating physician

  - Prolonged mechanical ventilation (> 7 days)
  - Initiation of renal replacement therapy
  - Prolonged (> 2 days) use of vasopressors such as norepinephrine or vasopressin
  - Renewed (restart after at least 2 days without) use of vasopressors such as norepinephrine or vasopressin
  - Other

- **Elaborate on other safety event** Source: Medical record or treating physician
- **Investigator assessment of safety event** Source: Investigator
- **Any other comments?**

18.2.18 End of study

- **What is the reason for the end of study?**

  - Completed follow up
  - Lost to follow up
  - Death
  - Discharge and no consent for follow up
  - Opt-out from routinely recorded (in-hospital) data collection and no consent for follow-up
  - Opt-out from routinely recorded (in-hospital) data collection and withdrawn consent for follow-up

- **Date of end of study**
- **Any other comments?**

19 References

1. Naghavi, M. *et al.* Global burden of 292 causes of death in 204 countries and territories and 660 subnational locations, 1990–2023: A systematic analysis for the Global Burden of Disease Study 2023. *The Lancet* **406**, 1811–1872 (2025).
2. GBD 2019 Diseases and Injuries Collaborators. Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: A systematic analysis for the global burden of disease study 2019. *The Lancet* **396**, 1204–1222 (2020).
3. Rauf, R. *et al.* Changes in the temporal distribution of in-hospital mortality in severely injured patients—an analysis of the TraumaRegister DGU. *PLOS ONE* **14**, e0212095 (2019).
4. Roy, N. *et al.* Learning from 2523 trauma deaths in india- opportunities to prevent in-hospital deaths. *BMC Health Serv Res* **17**, (2017).
5. Callcut, R. A. *et al.* The why and how our trauma patients die: A prospective multicenter western trauma association study. *J Trauma.* **86**, 864–870 (2019).
6. Ghorbani, P. *et al.* Analysis of preventable deaths and errors in trauma care in a scandinavian trauma level-i centre. *Acta Anaesthesiol Scand* **62**, 1146–1153 (2018).
7. Mohammad, A. *et al.* Educational and clinical impact of advanced trauma life support (ATLS) courses: A systematic review. *World J. Surg.* **38**, 322–329 (2013).
8. Jayaraman, S. *et al.* Advanced trauma life support training for hospital staff. *Cochrane Database Syst Rev* (2014).
9. Kadhum, M. *et al.* Are primary trauma care (PTC) courses beneficial in low- and middle-income countries - a systematic review. *Injury* **51**, 136–141 (2020).
10. Jin, J. *et al.* Effectiveness of quality improvement processes, interventions, and structure in trauma systems in low- and middle-income countries: A systematic review and meta-analysis. *World J. Surg.* **45**, 1982–1998 (2021).
11. McIver, R. *et al.* Effect of trauma quality improvement initiatives on outcomes and costs at community hospitals: A scoping review. *Injury* **55**, 111492 (2024).
12. Committee on Trauma. *Advanced trauma life support® student course manual*. (American College of Surgeons, 2018).
13. American College of Surgeons. *Resources for optimal care of the injured patient*. (American College of Surgeons, 2022).
14. Ali, J. *et al.* Demonstration of acquisition of trauma management skills by senior medical students completing the ATLS program. *J Trauma.* **38**, 687–691 (1995).
15. Ali, J. *et al.* Teaching effectiveness of the advanced trauma life support program as demonstrated by an objective structured clinical examination for practicing physicians. *World J. Surg.* **20**, 1121–1126 (1996).
16. Ali, J. *et al.* Comparison of performance of interns completing the old (1993) and new interactive (1997) advanced trauma life support courses. *J Trauma.* **46**, 80–86 (1999).
17. Putra, A. B. *et al.* Impact of Advanced Trauma Life Support Training for Improving Mortality Outcome: A Systematic Review and Meta-analysis. *The New Ropanasury Journal of Surgery* **8**, (2023).

18. Nakhid, Z. *et al.* Effect of trauma life support training on patient outcomes: A systematic review and meta-analysis. *Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine* (2026).
19. Vestrup, J. A. *et al.* Impact of advanced trauma life support training on early trauma management. *Am J Surg* **155**, 704–707 (1988).
20. Ariyanayagam, D. C. *et al.* The impact of the ATLS course on traffic accident mortality in trinidad and tobago. *West Indian Med J* **41**, 72–74 (1992).
21. Ali, J. *et al.* Trauma outcome improves following the advanced trauma life support program in a developing country. *J Trauma* **34**, 890–899 (1993).
22. Olden, G. D. J. van *et al.* Clinical impact of advanced trauma life support. *Am J Emerg Med* **22**, 522–525 (2004).
23. Wang, P. *et al.* Comparison of severe trauma care effect before and after advanced trauma life support training. *Chin J Traumatol* **13**, 341–344 (2010).
24. Noordin, S. *et al.* Evolution of trauma management at a tertiary care hospital: A cohort study. *International Journal of Surgery* **9**, 75–78 (2011).
25. Drimousis, P. G. *et al.* Advanced trauma life support certified physicians in a non trauma system setting: Is it enough? *Resuscitation* **82**, 180–184 (2011).
26. Hashmi, Z. G. *et al.* Hospital-based trauma quality improvement initiatives. *J Trauma* **75**, 60–68 (2013).
27. Petroze, R. T. *et al.* Can focused trauma education initiatives reduce mortality or improve resource utilization in a low-resource setting? *World J. Surg.* **39**, 926–933 (2014).
28. Giovanni Bellanova, R. B., Francesco Buccelletti. How formative courses about damage control surgery and non-operative management improved outcome and survival in unstable politrauma patients in a mountain trauma center. *Annali italiani di chirurgia* **87**, 68–74 (2016).
29. Magnone, S. *et al.* Impact of ATLS Guidelines and Trauma Team Introduction an 24 Hours Mortality in Severe Trauma in a Busy Italian Metropolitan Hospital: A Case Control Study. *Ulus Travma Derg* (2015).
30. Kamau, C. *et al.* Impact of advanced trauma life support training on 30-day mortality in severely injured patients at a Kenyan tertiary center: A retrospective matched case-control study. *International Journal of Emergency Medicine* **17**, 153 (2024).
31. Gerdin Wärnberg, M. *et al.* A pilot multicentre cluster randomised trial to compare the effect of trauma life support training programmes on patient and provider outcomes. *BMJ Open* **12**, e057504 (2022).
32. Gerdin Wärnberg, M. *et al.* Feasibility of a cluster randomised trial on the effect of trauma life support training: A pilot study in India. *BMJ Open* **15**, e099020 (2025).
33. David, S. *et al.* Patient-reported outcomes relevant to post-discharge trauma patients in urban India. (2024).
34. Hemming, K. *et al.* The stepped wedge cluster randomised trial: Rationale, design, analysis, and reporting. *BMJ* **350**, h391–h391 (2015).
35. Grantham, K. L. *et al.* The staircase cluster randomised trial design: A pragmatic alternative to the stepped wedge. *Statistical Methods in Medical Research* **33**, 24–41 (2024).

36. Hemming, K. *et al.* Reflection on modern methods: When is a stepped-wedge cluster randomized trial a good study design choice? *Int J Epidemiol* **49**, 1043–1052 (2020).
37. Kasza, J. *et al.* The batched stepped wedge design: A design robust to delays in cluster recruitment. *Stat Med* **41**, 3627–3641 (2022).
38. Harris, P. A. *et al.* Research electronic data capture (REDCap)–a metadata-driven methodology and workflow process for providing translational research informatics support. *Journal of Biomedical Informatics* **42**, 377–381 (2009).
39. Harris, P. A. *et al.* The REDCap consortium: Building an international community of software platform partners. *Journal of Biomedical Informatics* **95**, 103208 (2019).
40. Aukstakalnis, V. *et al.* Impact of video recordings review with structured debriefings on trauma team performance: A prospective observational cohort study. *European Journal of Trauma and Emergency Surgery: Official Publication of the European Trauma Society* (2024).
41. Nielsen, Niklas. Sedation, Temperature and Pressure after Cardiac Arrest and Resuscitation – the STEPCARE trial. (2023).
42. Diaz, T. *et al.* A call for standardised age-disaggregated health data. *The Lancet Healthy Longevity* **2**, e436–e443 (2021).
43. Li, F. *et al.* Mixed-effects models for the design and analysis of stepped wedge cluster randomized trials: An overview. *Stat Methods Med Res* **30**, 612–639 (2020).
44. Hemming, K. *et al.* A tutorial on sample size calculation for multiple-period cluster randomized parallel, cross-over and stepped-wedge trials using the shiny CRT calculator. *Int J Epidemiol* **49**, 979–995 (2020).
45. Campbell, M. K. *et al.* Determinants of the intracluster correlation coefficient in cluster randomized trials: The case of implementation research. *Clinical Trials* **2**, 99–107 (2005).
46. Eldridge, S. M. *et al.* How big should the pilot study for my cluster randomised trial be? *Stat Methods Med Res* **25**, 1039–1056 (2015).
47. Martin, J. *et al.* Intra-cluster and inter-period correlation coefficients for cross-sectional cluster randomised controlled trials for type-2 diabetes in UK primary care. *Trials* **17**, (2016).
48. Korevaar, E. *et al.* Intra-cluster correlations from the CLustered OUtcome dataset bank to inform the design of longitudinal cluster trials. *Clinical Trials* **18**, 529–540 (2021).
49. Lashoher, A. *et al.* Implementation of the World Health Organization Trauma Care Checklist Program in 11 Centers Across Multiple Economic Strata: Effect on Care Process Measures. *World Journal of Surgery* **41**, 954–962 (2017).
50. Kapitan, E. *et al.* The effect of trauma quality improvement programme implementation on quality of life among trauma patients in urban India. *medRxiv* 2024.09.28.24314529 (2024).
51. Higgins, A. M. *et al.* The psychometric properties and minimal clinically important difference for disability assessment using WHODAS 2.0 in critically ill patients. *Critical Care and Resuscitation* **23**, 103–112 (2021).
52. ICMJE | Recommendations | Defining the Role of Authors and Contributors.

20 Appendices

20.1 Initial hospital screening instrument

Screening call

This form is for screening potentially eligible clusters for the ATLS vs standard care trial. Please fill it in while talking to the hospital representative. Thank you so much for helping with this task!

Please complete the questions below.

Synopsis Title Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Cluster Randomised Trial

Rationale Trauma is a massive global health issue. Many training programmes have been developed to help physicians in the initial management of trauma patients. Advanced Trauma Life Support® (ATLS®) is the most popular of these programmes and have been used to train over one million physicians worldwide. Despite its widespread use, there are no controlled trials showing that ATLS® improves patient outcomes. Multiple systematic reviews emphasise the need for such trials.

Aim To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

Primary Outcome In-hospital mortality within 30 days of arrival at the emergency department.

Trial Design Batched stepped-wedge cluster randomised trial in India.

Trial Population Adult trauma patients presenting to the emergency department of a participating hospital.

Sample Size 30 clusters and 4320 patients.

Eligibility Criteria

Cluster will be hospitals with a baseline admission rate of at least 400 patients with trauma per year or 35 patients with trauma per month for at least the last six months, that provide emergency surgical and orthopaedic services around the clock, and where no more than 25% of initial trauma care providers trained in a formalised trauma life support training programme.

Patients will be at least 15 years old, who present to the emergency department of participating hospitals with a history of trauma occuring less than 48 hours before arrival, and who are admitted or die between and admission, or who are transferred from the emergency department of a participating hospital to another hospital for admission.

Intervention The intervention will be ATLS® training, a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. Physicians will be trained in an accredited ATLS® training facility in India.

Ethical Considerations In-hospital data collection will be conducted under a waiver of informed consent. Patients will be informed about the trial and their right to opt out of data collection. Patients will be informed that they can withdraw their data from the trial at any time.

Trial Period 2024-10-01 to 2029-10-01

Hospital details

Hospital name and address

Eligibility assessment	
Does the hospital admit trauma patients?	◯ Yes ◯ No
Is the hospital representative interested in potentially participating?	◯ Yes ◯ No (Comment: _____)
If the hospital representative is not interested in participating then you can go ahead and submit the form.	
How many trauma patients aged 15 years or older, excluding patients with isolated limb injuries, are admitted each month?	◯ < 30 ◯ 30-60 ◯ > 60 ◯ Not sure (Comment: _____)
Does the hospital provide emergency surgery and orthopaedic services around the clock?	◯ Yes ◯ No (Comment: _____)
Out of the physicians involved in the initial resuscitation of trauma patients, are less than 25% trained in a formalised trauma life support training programme like ATLS or Primary Trauma Care?	◯ Yes ◯ No (Comment (like name of other training programme): _____)
Unfortunately, the hospital does not fulfil the eligibility criteria and you may go ahead and submit the form.	
The hospital fulfills the cluster eligibility criteria. Please enter the hospital representative's contact details: Name E-mail Phone number _____ _____ _____ Before you can proceed to fill in descriptive information about the potential cluster, please confirm that the contact details are complete: _____	

Cluster descriptive information	
Out of the patients with trauma who present to the emergency department, what percentage are referrals or transfers from other hospitals?	◯ < 20% ◯ 20-50% ◯ 51-80% ◯ > 80% ◯ Not sure (Comment: _____)
Out of the patients with trauma who are admitted, what percentage are transferred to other hospitals?	◯ < 10% ◯ 10-20% ◯ 21-30% ◯ >30% ◯ Not sure (Comment: _____)

Who performs the initial resuscitation of trauma patients as they arrive to the hospital?	☐ Casualty medical officers ☐ Emergency medicine residents ☐ Surgical residents ☐ Not sure (Comment: _____)
How many CMOs work in the emergency department? This question is here to help us estimate the number of people we will need to train.	◯ < 10 ◯ 10-20 ◯ 21-30 ◯ > 30 ◯ Not sure (Comment: _____)
How many emergency medicine/general surgery residents are admitted each year? This question is here to help us estimate the number of people we will need to train.	◯ < 10 ◯ 10-20 ◯ 21-30 ◯ > 30 ◯ Not sure (Comment: _____)
What specialities are available around the clock to care for trauma patients?	☐ General surgery ☐ Orthopaedics ☐ Neurosurgery ☐ Vascular surgery ☐ Interventional radiology ☐ Emergency medicine ☐ Not sure (Comment: _____)
What facilities are available around the clock?	☐ X-ray ☐ Ultrasound/FAST ☐ CT ☐ MRI ☐ Blood bank ☐ Not sure (Comment: _____)
How many beds does the hospital have?	◯ < 250 ◯ 250-500 ◯ 501-750 ◯ 751-1000 ◯ >1000 ◯ Not sure (Comment: _____)
How many ICU beds does the hospital have?	◯ No ICU beds ◯ 1-10 ◯ 11-20 ◯ 21-30 ◯ > 30 ◯ Not sure (Comment: _____)
How many dedicated trauma beds does the hospital have?	◯ No dedicated trauma beds ◯ 1-10 ◯ 11-20 ◯ 21-30 ◯ > 30 ◯ Not sure (Comment: _____)

Please submit the form once you have completed all questions above.

20.2 In-depth hospital screening interview instrument

Hospital screening interview for the ATLS vs standard care trial

This is the screening interview form for the Advanced Trauma Life Support® vs Standard Care trial planned by Karolinska Institutet along with The George Institute. You have expressed preliminary interest inparticipating in this trial. We are undertaking this hospital screening interview inorder to assess whether the study could be conducted at your hospital. We appreciate your efforts to answer as many of these questions as possible and we will follow up on your responses and any questions you may have in a separate call. Thank you!

Synopsis Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Cluster Randomised Trial Rationale Trauma is a massive global health issue. Many training programmes have been developed to help physicians in the initial management of trauma patients. Among these programmes, Advanced Trauma Life Support® (ATLS®) is the most popular, having trained over one million physicians worldwide. Despite its widespread use, there are no controlled trials showing that ATLS® improves patient outcomes. Multiple systematic reviews emphasise the need for such trials.

Aim To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

Primary Outcome All-cause mortality within 30 days of arrival at the emergency department.

Trial Design Batched stepped-wedge cluster randomised trial in India.

Trial Population Adult trauma patients presenting to the emergency department of a participating hospital.

Sample Size 30 clusters and 4320 patients.

Eligibility Criteria

Hospitals are secondary or tertiary hospitals in India that admit or refer/transfer for admission at least 400 patients with trauma per year.

Clusters are one or more units of physicians providing initial trauma care in the emergency department of tertiary hospitals in India.

Patients participants are adult trauma patients who presents to the emergency department of participating hospitals and are admitted or transferred for admission.

Intervention The intervention will be ATLS® training, a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. Physicians will be trained in an accredited ATLS® training facility in India.

Ethical Considerations We will use an opt-out consent approach, in which consent is presumed unless actively declined. Note that consent here refers to consent to data collection, as it will not be possible for patients to opt out from being subjected to the intervention. This approach is justified because the trial can be considered to involve only minimal risk and the data

collection is non-invasive and mostly involve extracting routinely collected data from medical records. Patient participants will be informed about the study and their right to opt out once they are admitted or telephonically if they are transferred. Patients will be informed that they can withdraw their data from the trial at any time before final analysis of the data.

Trial Period October 1, 2024, to September 30, 2029

Hospital details

[hospital_address]

Contact person details

Name [contact_name]
E-mail [contact_email]
Phone number [contact_phone_number]

Will you [contact_name] also be the site investigator?	◯ Yes ◯ No

Investigator details

Please enter the name and contact details of the site investigator

Name _____
E-mail _____
Phone number _____

Please enter these additional investigator details

Designation _____
Specialization _____
State Medical Council registration number _____

Is the investigator trained in International Council for Harmonisation, Guideline for Good Clinical Practice (ICH GCP)?	◯ Yes ◯ No
Will there be a co-investigator at your site?	◯ Yes ◯ No
Will you [contact_name] be the co-investigator?	◯ Yes ◯ No

Please enter the name and contact details of the site investigator

Name _____
E-mail _____
Phone number _____

Please enter these additional co-investigator details Designation _____ Specialization _____ State Medical Council registration number _____	
Is the co-investigator trained in International Council for Harmonisation, Guideline for Good Clinical Practice (ICH GCP)?	◯ Yes ◯ No

Ethical review details

Does your hospital have an ethics committee registered with CDSCO?	◯ Yes ◯ No
Please enter the ethics committee registration number	_____
In the next three months when is your IEC meeting up?	_____
What is the expected timeline for ethics review at your site?	_____
In which languages do you think the consent form should be translated, considering the languages spoken by the potential participants treated at your hospital?	_____

Departmental logistics

Does your hospital require any additional departmental review besides the ethics? Can you please elaborate on that review?	_____
Are there any potential logistical issues which may interfere with set up or running ofthis project at your site?	_____ (E.g. contract review, adequate space, lack of resources etc.)
What is the expected timeline for contract review, negotiations and execution (in days)?	_____
What is the maximum expected timeline?	_____

Initial trauma care	
How do patients typically arrive to your hospital?	
Who are involved in the management of trauma patients in the emergency department?	
What happens when additional expertise is needed?	
What is the role of the casualty medical officers?	
Are physicians organised in units?	
How big are those units?	
How are those units composed in terms of residents and faculty?	
How often do the units rotate?	
How many units are there working in the emergency department?	
How many trauma patients aged 15 years or older are admitted per day, excluding patients with isolated limb injuries and those who are admitted directly to the ward?	

Intervention and patient inclusion	
How many patients do you think you could include in to the proposed trial per month? (We need to include at least 12 patients per month)	
What is the basis of your patient enrollment estimate?	(For example database review, emergency department record, review of patient records, other)
Do you see any problems with including 12 patients per month at your site? Can you please elaborate on those problems?	

All hospitals in this trial will receive the intervention. The intervention is that we will train approximately 10 physicians providing initial trauma care in ATLS in your hospital. Who do you think we should train to maximise the effect?	_____ (Surgical residents? Emergency medicine residents? Casualty medical officers? Someone else??)
The time point when the training will be implemented will be randomised, but there will be a minimum of three months between the start of the data collection and the training. The training will happen during a one month long "transition period". How long notice do you need to plan the participation of the physicians from your hospital?	_____
Are you aware of any plans to train providers in any formalised trauma life support training programme during the next few years?	_____
Are you aware of any plans to implement other interventions or changes that may radically change how you treat trauma patients at your site?	_____ (For example building a trauma centre, building a new emergency department, shifting the CT)
If we would like to visit your hospital to observe trauma care delivery in the emergency department and talk to providers, how can that be arranged?	_____
General	
How are the patient medical records organised at your site?	☐ Hard-copy ☐ Electronic ☐ Not sure
Do you currently have any competing studies or are you committed to new competing studies?	_____
Do you have access to a computer with high-speed internet access?	◯ Yes ◯ No
Do you currently have the necessary study team including research coordinator and co-investigators to conduct this study? Can you please elaborate on the composition and experience of that team?	_____
What are your expectations of this trial?	_____
Do you have any questions or comments regarding this trial?	_____
Are you interested in participating in this trial?	◯ Yes ◯ No

If you have any questions, feel free to contact Martin Gerdin Wärnberg (martin.gerdin@ki.se), Monty Khajanchi (monta32@gmail.com) or Samriddhi Ranjan (sranjan@georgeinstitute.org.in)

20.3 Case Record Form

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Screening V1.0.01.10.24

Screening ID	_____
1. Date of screening	_____
2. Date of data entry	_____

Inclusion criteria	
1. Is the patient at least 15 years old?	(Source: Medical record or interview)
2. Did the patient present with a history of trauma defined as having any of the reasons listed in the International Classification of Diseases chapter XX as the reason for presenting? Please see https://icd.who.int/browse10/2019/en#/XX for a complete list of ICD-10 codes	(Source: Medical record or interview)
3. Did the trauma occur less than 48 hours before arrival to the hospital?	(Source: Medical record or interview)
4. Was the patient admitted?	(Source: Medical record)
5. Did the patient die after arrival but before admission?	(Source: Medical record)
6. Was the patient transferred to another hospital for admission?	(Source: Medical record)

Exclusion criteria	
1. Did the patient present with isolated closed extremity fracture?	(Source: Medical record)
2. Was the patient directly admitted to a ward without being seen by a physician in the emergency department?	(Source: Medical record)

Eligibility	
The patient is not eligible for inclusion.	
Is the patient eligible? (Eligible=1; Not Eligible=0)	_____
Any other comments?	_____

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Consent V1.0.01.10.24

Study Consent

In this trial, consent refers to consent for data collection. It is not possible for patients to opt out from being subjected to the intervention, as the intervention is delivered at the cluster level. Patient participants will be included in this trial under the following modes of consent:
• Opt-out consent for collection of routinely recorded data
• Opt-in consent and assent for non-routinely recorded data, including but not restricted to Quality of Life (EQ5D5L), Disability (WHODAS 2.0) and Return to Work.
• Waiver of informed consent for patients who are unconscious or otherwise unable to provide consent and do not have a legally acceptable representative.
When possible, all patient participants must be approached and provided with information about the study, the option to opt out, and consent for collection of non-routinely recorded data.

Section I: Consent Wavier

Please note that the consent for the collection of the routinely recorded data (in-hospital) will be presumed unless actively declined by the participant/ legally acceptable representative (LAR), using the opt-out form. Information for all forms except for baseline characteristics (marital and work status, education and income), follow-up (Quality of Life (EQ5D5L), Disability (WHODAS 2.0) and Return to Work) will be presumed, unless opted-out.

1. Is this patient included under the waiver of informed consent because the patient is unconscious or otherwise unable to provide consent and do not have a legally acceptable representative?	

Section II: Opt in consent for follow up data collection

1. Did the participant/ or legally acceptable representative (LAR) provided consent for collection of non-routinely recorded data	
2. Who gave consent for collection of non-routinely recorded data?	
3. What is the LAR's relationship with the participant?	_____
4. Why was Legally acceptable representative (LAR) approached for consent for collection of non-routinely recorded data?	
5. Date when participant or legally acceptable representative (LAR) gave consent for collection of non-routinely recorded data.	_____
6. How did the participant or legally acceptable representative (LAR) consent for collection of non-routinely recorded data?	
7. Reconsenting done?	

7.1 Date when the participant was reconsented?	_____
7.2 Why consenting was not done?	_____

Section III: Assent form	
1. Did the minor give assent for collection of non-routinely recorded data?	
2. Date when the minor gave assent for collection of non-routinely recorded data.	_____
3. In case the minor refused to participate, date when minor refused	_____

Section IV: Opt out form	
1. Is the participant or LAR wants to opt out from study?	
2. Who opted-out of the routinely recorded data (in-hospital)?	
3. Date when participant or legally acceptable representative (LAR) opted-out.	_____
4. Did the participant or legally acceptable representative (LAR) suggested to delete all the previously recorded data?	
Any other comments?	_____

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Consent_Withdrawn V1.0.01.10.24

Consent withdrawal	
1. Does the participant or legally acceptable representative (LAR) want to withdraw the consent?	
2. Date of consent withdrawal for follow-up data collection.	_____
3. Procedure(s) for which consent has been withdrawn	
Any other comments?	_____

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Baseline V1.0.01.10.24

1. Age in years	_____ (Source: Medical record of interview)
2. Sex	(Source: Medical record of interview)
3. Current marital status	(Requires opt-in consent, not routinely recorded. Source: Interview)
4. Education level	(Requires opt-in consent, not routinely recorded. Source: Interview)
5. If other, please specify	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)
6. Main work status	(Requires opt-in consent, not routinely recorded. Source: Interview)
7. If other, please specify	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)
8. Income level in INR per month	(Requires opt-in consent, not routinely recorded. Source: Interview)
9. Mechanism of injury description	_____
10. Mechanism of injury ICD 10 code	(Coded using ICD 10. Please see https://icd.who.int/browse10/2019/en#/XX for a complete list of ICD-10 codes. Source: Medical record)

11. Clinical Frailty Scale Very Fit - People who are robust, active, energetic and motivated. These people commonly exercise regularly. They are among the fittest for their age. Fit - People who have no active disease symptoms but are less fit than category 1. Often, they exercise or are very active occasionally, e.g. seasonally. Managing well - People whose medical problems are well controlled, but are not regularly active beyond routine walking. Living with very mild frailty - While not dependent on others for daily help, often symptoms limit activities. A common complaint is being "slowed up", and/or being tired during the day. Living with mild frailty - These people often have more evident slowing, and need help in high order IADLs (finances, transportation, heavy housework, medications). Typically, mild frailty progressively impairs shopping and walking outside alone, meal preparation and housework. Living with moderate frailty - People need help with all outside activities and with keeping house. Inside, they often have problems with stairs and need help with bathing and might need minimal assistance (cuing, standby) with dressing. Living with severe frailty - Completely dependent for personal care, from whatever cause (physical or cognitive). Even so, they seem stable and not at high risk of dying (within ~ 6 months). Living with very severe frailty - Completely dependent, approaching the end of life. Typically, they could not recover even from a minor illness. Terminally ill - Approaching the end of life. This category applies to people with a life expectancy < 6 months, who are not otherwise evidently frail.	(Source: Medical record or treating physician)

12. Comorbidities (Charlson Comorbidity Index) Myocardial infarction - History of definite or probable MI (EKG changes and/or enzyme changes) Congestive heart failure - Exertional or paroxysmal nocturnal dyspnea and has responded to digitalis, diuretics, or afterload reducing agents Peripheral vascular disease - Intermittent claudication or past bypass for chronic arterial insufficiency, history of gangrene or acute arterial insufficiency, or untreated thoracic or abdominal aneurysm (≥6 cm) Cerebrovascular disease - History of a cerebrovascular accident with minor or no residua and transient ischemic attacks Dementia - Chronic cognitive deficit Chronic pulmonary disease - Asthma, chronic bronchitis, emphysema, and other lung disease who have ongoing symptoms such as dyspnea or cough, with mild or moderate activity. Peptic ulcer disease - Any history of treatment for ulcer disease or history of ulcer bleeding Liver disease - Severe = cirrhosis and portal hypertension with variceal bleeding history, moderate = cirrhosis and portal hypertension but no variceal bleeding history, mild = chronic hepatitis (or cirrhosis without portal hypertension) Renal disease - Severe = on dialysis, status post kidney transplant, uremia, moderate = creatinine >3 mg/dL (0.27 mmol/L)	(Source: Medical record, treating physician or interview)
12.1 Please specify	_____
13. Severity of liver disease	(Source: Medical record, treating physician or interview)
14. Severity of diabetes	(Source: Medical record, treating physician or interview)
15. Severity of malignancy	(Source: Medical record, treating physician or interview)
Any other comments?	_____

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Prehospital V1.0.01.10.24

1. Date of injury	_____ (Source: Medical record of interview)
1.1 Time of injury	_____
2. Mode of transport to the participating hospital	(Source: Medical record of interview)
2.1 If other, please specify	_____ (Source: Medical record of interview)
3. Referred or transferred to the participating hospital from another hospital	(Source: Medical record of interview)
4. Date of referring or transferring	_____
5. Time of referring or transferring	_____
Any other comments?	_____

ATLS adherence V1.1.22.10.24

ATLS adherence checklist

Airway	
1. Airway patency checked	(Source: Observation)
Breathing	
1. Chest wall palpated	(Source: Observation)
2. Breath sounds checked	(Source: Observation)
3. Respiratory rate measured	(Source: Observation)
4. Saturation (SpO2) measured	(Source: Observation)
Circulation	
1. Heart rate measured	(Source: Observation)
2. Blood pressure measured	(Source: Observation)
3. Abdomen palpated	(Source: Observation)
4. Thighs palpated	(Source: Observation)
5. IV access obtained	(Source: Observation)
Disability	
1. GCS checked	(Source: Observation)
2. Pupils checked	(Source: Observation)
Exposure	
1. Patients exposed for assessment	
2. Temperature measured	(Source: Observation)
3. Interventions and adjuncts performed according to ATLS	

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Airway interventions	
1. Which airway interventions were performed?	(Source: Observation)
2. If other airway interventions given, specify	_____
3. Were airway interventions performed while minimising c-spine movement?	(Source: Observation)
Breathing interventions	
1. Which breathing interventions were performed?	(Source: Observation)
2. If other breathing Interventions done, specify	_____
Circulation interventions	
1. Which circulation interventions and adjuncts were performed?	(Source: Observation)
2. If other circulation Interventions done, specify	_____
Disability interventions	
1. Which disability intervention was performed?	(Source: Observation)
2. If other disability interventions done, specify	_____
Exposure interventions	
1. Which exposure intervention was performed?	(Source: Observation)
2. If other exposure interventions done, specify	_____
Any other comments?	_____

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Emergency Department V1.0.01.10.24

1. Date of arrival to the emergency department at the participating hospital.	_____ (Source: Medical record of interview)
2 Time of arrival to the emergency department at the participating hospital	_____
3. First recorded systolic blood pressure (mmHg)	_____ (Source: Medical record)
4. First recorded diastolic blood pressure (mmHg)	_____ (Source: Medical record)
5. First recorded heart rate (beats per minute)	_____ (Source: Medical record)
6. First recorded respiratory rate (breaths per minute)	_____ (Source: Medical record)
7. First recorded Glasgow Coma Scale	_____ (Source: Medical record)
8. First recorded body temperature (°F)	_____ (Source: Medical record)
9. First recorded oxygen saturation (%)	_____ (Source: Medical record)
10. Emergency department disposition	(Source: Medical record)
10.1 Date of death	_____
10.2 Time of death	_____
11. If other, please specify	_____ (Source: Medical record)
12. Date and time of referral or transfer for admission	_____ (Source: Medical record)
Any other comments?	_____

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Hospital V1.0.01.10.24

1. Date of admission to the participating hospital	_____ (Source: Medical record)
2 Time of admission to the participating hospital	_____ (Source: Medical record)
3. Type of admitting ward	(Source: Medical record)
4. Ward name or number	_____ (Source: Medical record)
5. Admitted to intensive care unit during admission	(Source: Medical record)
6. Date of first intensive care unit admission	_____ (Source: Medical record)
7. Time of first intensive care unit admission	_____ (Source: Medical record)
8. Was the participant discharged from intensive care unit?	
9. Date of first intensive care unit discharge	_____ (Source: Medical record)
10. Time of first intensive care unit discharge	_____ (Source: Medical record)
11. Hospital disposition	(Source: Medical record)
12. Was the patient transferred to another hospital for admission?	(Source: Medical record)
13. Date of discharge or transfer from participating hospital	_____ (Source: Medical record)
14. Time of discharge or transfer from participating hospital	_____ (Source: Medical record)
Any other comments?	_____

Surgery V1.0.01.10.24

1. Surgery done?	
2. Date of surgical procedure	_____ (A surgical procedure is defined as any procedure performed in the operating room, interventional dropdownlogy suite, or at the bedside, requiring general or regional anesthesia. Source: Medical record)
3. Time of surgical procedure	_____ (A surgical procedure is defined as any procedure performed in the operating room, interventional dropdownlogy suite, or at the bedside, requiring general or regional anesthesia. Source: Medical record)
4. Preoperative ASA score	(Source: Medical record or treating physician)
5. Description of procedure	_____ (Source: Medical record)
6. Procedure coded according to SNOMED CT	_____ (Source: Medical record)
Any other comments?	_____

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Imaging V1.0.01.10.24

1. Date and time of imaging	_____ (Source: Medical record)
2 Time of imaging	_____ (Source: Medical record)
3. Type of imaging	(Source: Medical record)
Any other comments?	_____

Transfusion V1.0.01.10.24

1. Transfusion done?	
2. Date of transfusion	_____ (Source: Medical record)
3. Time of transfusion	_____ (Source: Medical record)
4. Type of blood product	(Source: Medical record)
5 Other specify	_____
6. Number of units transfused	_____ (Source: Medical record)

Any other comments?

_____

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Injury V1.0.01.10.24

1. Injury description	_____ (Source: Medical record)
2. Injury ICD 10 code	(Coded using ICD 10. Source: Medical record)
3. Injury source data	(Source: Medical record)
4. AIS code	_____
Any other comments?	_____

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Individual Mortality Status V1.0.01.10.24

1. Is the patient dead?	(Source: Medical record or interview)
2. Date of death	_____ (Source: Medical record or interview)
3. Time of death	_____
Any other comments?	_____

Quality of Life (EQ5D5L)

Health Questionnaire

English version

VERSION FOR INTERVIEWER ADMINISTRATION

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Note to interviewer: although allowance should be made for the interviewer's particular style of speaking, the wording of the questionnaire instructions should be followed as closely as possible. In the case of the EQ-5D-5L descriptive system of the questionnaire, the precise wording must be followed.

If the respondent has difficulty choosing a response or asks for clarification, the interviewer should repeat the question word for word and ask the respondent to answer in a way that most closely resembles his or her thoughts about his or her health today.

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INTRODUCTION

(Note to interviewer: please read the following to the respondent.)

We are trying to find out what you think about your health. I will explain what to do as I go along, but please interrupt me if you do not understand something or if things are not clear to you. There are no right or wrong answers. We are interested only in your personal view.

First, I am going to read out some questions. Each question has a choice of five answers. Please tell me which answer best describes your health TODAY.

Do not choose more than one answer in each group of questions.

(Note to interviewer: first read all five options for each question. Then ask the respondent to choose which one applies to him/herself. Repeat the question and options if necessary. Mark the appropriate box under each heading. You may need to remind the respondent regularly that the timeframe is TODAY.)

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EQ-5D DESCRIPTIVE SYSTEM

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1. Date of filling this form

_____

2. First, I would like to ask you about MOBILITY. Would you say that:
(Requires opt-in consent, not routinely recorded. Source: Interview)

◯ You have no problems in walking about?
◯ You have slight problems in walking about?
◯ You have moderate problems in walking about?
◯ You have severe problems in walking about?
◯ You are unable to walk about?

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3. Next, I would like to ask you about SELF-CARE. Would you say that:
(Requires opt-in consent, not routinely recorded. Source: Interview)

◯ You have no problems washing or dressing yourself?
◯ You have slight problems washing or dressing yourself?
◯ You have moderate problems washing or dressing yourself?
◯ You have severe problems washing or dressing yourself?
◯ You are unable to wash or dress yourself?

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4. Next, I would like to ask you about USUAL ACTIVITIES, for example, work, study, housework, family or leisure activities. Would you say that:
(Requires opt-in consent, not routinely recorded. Source: Interview)

◯ You have no problems doing your usual activities?
◯ You have slight problems doing your usual activities?
◯ You have moderate problems doing your usual activities?
◯ You have severe problems doing your usual activities?
◯ You are unable to do your usual activities?

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5. Next, I would like to ask you about PAIN OR DISCOMFORT. Would you say that:
(Requires opt-in consent, not routinely recorded. Source: Interview)

◯ You have no pain or discomfort?
◯ You have slight pain or discomfort?
◯ You have moderate pain or discomfort?
◯ You have severe pain or discomfort?
◯ You have extreme pain or discomfort?

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6. Finally, I would like to ask you about ANXIETY OR DEPRESSION. Would you say that:
(Requires opt-in consent, not routinely recorded. Source: Interview)

◯ You are not anxious or depressed?
◯ You are slightly anxious or depressed?
◯ You are moderately anxious or depressed?
◯ You are severely anxious or depressed?
◯ You are extremely anxious or depressed?

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EQ-5D VAS

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- Now, I would like to ask you to say how good or bad your health is TODAY.

- I would like you to picture in your mind a vertical line that is numbered from 0 to 100.
(Note to interviewer: if interviewing face-to-face, please show the respondent the VAS line.)

- 100 at the top of the line means the best health you can imagine.
0 at the bottom of the line means the worst health you can imagine.

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- I would now like you to tell me the point on this line where you would put your health TODAY. (Note to interviewer: mark the line at the point indicating the respondent's health today.)	0 - The worst health you can imagine	50	100 - The best health you can imagine

(Place a mark on the scale above)

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Any other comments?

Disability (WHODAS 2.0)

Date of form filling	_____
1. Who are you interviewing?	(Requires opt-in consent, not routinely recorded. Source: Interview)
2. What is the relationship between the representative and the participant?	(Requires opt-in consent, not routinely recorded. Source: Interview)
3. If other, please specify	_____

Instructions to the interviewer are written in bold - do not read these aloud.

Text for the respondent to hear is written in italic print in blue. Read this text aloud.

Say to respondent:
The interview is about difficulties people have because of health conditions.

By health condition I mean diseases or illnesses, or other health problems that may be short or long lasting; injuries; mental or emotional problems; and problems with alcohol or drugs.

Remember to keep all of your health problems in mind as you answer the questions. When I ask you about difficulties in doing an activity think about...

- Increased effort
- Discomfort or pain
- Slowness
- Changes in the way you do the activity When answering, I'd like you to think back over the past 30 days. I would also like you to answer these questions thinking about how much difficulty you have had, on average, over the past 30 days, while doing the activity as you usually do it.

Use this scale when responding: None, mild, moderate, severe, extreme or cannot do.

In the past 30 days, how much difficulty did you have in:

1. Standing for long periods such as 30 minutes?	(Requires opt-in consent, not routinely recorded. Source: Interview)
2. Taking care of your household responsibilities?	(Requires opt-in consent, not routinely recorded. Source: Interview)
3. Learning a new task, for example, learning how to get to a new place?	(Requires opt-in consent, not routinely recorded. Source: Interview)

4. How much of a problem did you have joining in community activities (for example, festivities, religious or other activities) in the same way as anyone else can?	(Requires opt-in consent, not routinely recorded. Source: Interview)
5. How much have you been emotionally affected by your health problems?	(Requires opt-in consent, not routinely recorded. Source: Interview)

In the past 30 days, how much difficulty did you have in:	
1. Concentrating on doing something for ten minutes?	(Requires opt-in consent, not routinely recorded. Source: Interview)
2. Walking a long distance such as a kilometre [or equivalent]?	(Requires opt-in consent, not routinely recorded. Source: Interview)
3. Washing your whole body?	(Requires opt-in consent, not routinely recorded. Source: Interview)
4. Getting dressed?	(Requires opt-in consent, not routinely recorded. Source: Interview)
5. Dealing with people you do not know?	(Requires opt-in consent, not routinely recorded. Source: Interview)
6. Maintaining a friendship?	(Requires opt-in consent, not routinely recorded. Source: Interview)
7. Your day-to-day work/school?	(Requires opt-in consent, not routinely recorded. Source: Interview)

Instructions to the interviewer are written in bold - do not read these aloud.

Text for the respondent to hear is written in italic print in blue. Read this text aloud.

Say to respondent:
The interview is about difficulties people have because of health conditions.

By health condition I mean diseases or illnesses, or other health problems that may be short or long lasting; injuries; mental or emotional problems; and problems with alcohol or drugs.

Remember to keep all of your health problems in mind as you answer the questions. When I ask you about difficulties in doing an activity think about...

- Increased effort
- Discomfort or pain
- Slowness
- Changes in the way you do the activity When answering, I'd like you to think back over the past 30 days and, to the best of your knowledge, answer these questions thinking about how much difficulty your friend, relative or carer had while doing the following activities. I will use

the term "relative" to mean "friend", "relative" or "carer". For each question, please give only one response. In the past 30 days, how much difficulty did your relative have in:	
1. Standing for long periods such as 30 minutes?	(Requires opt-in consent, not routinely recorded. Source: Interview)
2. Taking care of his or her household responsibilities?	(Requires opt-in consent, not routinely recorded. Source: Interview)
3. Learning a new task, for example, learning how to get to a new place?	(Requires opt-in consent, not routinely recorded. Source: Interview)
4. How much of a problem did he or she have joining in community activities (for example, festivities, religious or other activities) in the same way as anyone else can?	(Requires opt-in consent, not routinely recorded. Source: Interview)
5. How much has your relative been emotionally affected by his or her health condition?	(Requires opt-in consent, not routinely recorded. Source: Interview)
In the past 30 days, how much difficulty did your relative have in:	
1. Concentrating on doing something for ten minutes?	(Requires opt-in consent, not routinely recorded. Source: Interview)
2. Walking a long distance such as a kilometre [or equivalent]?	(Requires opt-in consent, not routinely recorded. Source: Interview)
3. Washing his or her whole body?	(Requires opt-in consent, not routinely recorded. Source: Interview)
4. Getting dressed?	(Requires opt-in consent, not routinely recorded. Source: Interview)
5. Dealing with people he or she does not know?	(Requires opt-in consent, not routinely recorded. Source: Interview)
6. Maintaining a friendship?	(Requires opt-in consent, not routinely recorded. Source: Interview)
7. His or her day-to-day work/school?	(Requires opt-in consent, not routinely recorded. Source: Interview)
Number of days	
1. Overall, in the past 30 days, how many days were these difficulties present?	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)
2. In the past 30 days, for how many days were you totally unable to carry out your usual activities or work because of any health condition?	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)

3. In the past 30 days, not counting the days that you were totally unable, for how many days did you cut back or reduce your usual activities or work because of any health condition?	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)
Any other comments?	_____

Confidential

Return To Work V1.0.01.10.24

1. Date of form filling	_____
2. Did the participant return to work?	
3. Date and time of return to work	_____ (Requires opt-in consent, not routinely recorded. Source: Interview)
4. Work status	(Requires opt-in consent, not routinely recorded. Source: Interview)
Any other comments?	_____

Confidential

Safety Events V1.0.01.10.24

1. Is any safety event reported?	
2. Date reported to trial management team of safety event	_____
3. Type of safety event	(Source: Medical record or treating physician)
4. Elaborate on other safety event	_____ (Source: Medical record or treating physician)
5. Investigator assessment of safety event	_____ (Source: Investigator)
Any other comments?	_____

Confidential

End Of Study V1.0.01.10.24

1. What is the reason for the end of study?

2. Date of end of study

_____

Any other comments?

_____

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Scheduling

Day 7

_____

Day 30

_____

Day 90

_____

Scheduling Page

Event Date
Date of arrival to the emergency departments [initial_assessment_arm_1][arrival_datetime]
Day 7 [day7]
Day 30 [day30]
Day 90 [day90]

Review

Status

Reviewed by

Review comments

(For example, comments about main quality issues, if any.)